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Statistical Methods for Non-adherence in Non-inferiority Trials: Useful and Used? A Systematic Review

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STATISTICAL METHODS FOR NON-ADHERENCE IN NON-INFERIORITY TRIALS: USEFUL AND USED? A SYSTEMATIC REVIEW

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ABSTRACT

Background: In non-inferiority trials with non-adherence to interventions (or non-compliance), intention-to-treat and per-protocol analyses are often performed; however non-random non-adherence biases estimates of efficacy.

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Objective: To identify statistical methods that adjust for the impact of non-adherence and thus estimate the causal effects of experimental interventions in non-inferiority trials.

Design: A systematic review was conducted by searching the Ovid MEDLINE® database (31st December 2020) to identify 1) randomised trials with a primary analysis for non-inferiority that applied (or planned to apply) statistical methods to account for the impact of non-adherence to interventions, and 2) methodology papers that described such statistical methods and included a non-inferiority trial application.

Outcomes: The statistical methods identified, their impacts on non-inferiority conclusions, and their advantages/disadvantages.

Results: A total of 24 papers were included (4 protocols, 13 results papers, 7 methodology papers) reporting relevant methods on 26 occasions. The most common were instrumental variable approaches (n=9), including observed adherence as a covariate within a regression model (n=3), and modelling adherence as a time-varying covariate in a time-to-event analysis (n=3). Other methods included rank preserving structural failure time models and inverse-probability-of-treatment weighting. The methods identified in protocols and results papers were more commonly specified as sensitivity analyses (n=13) than primary analyses (n=3). Twelve results papers included an alternative analysis of the same outcome; conclusions regarding non-inferiority were in agreement on six occasions and could not be compared on six occasions (different measures of effect or results not provided in full).

Conclusions: Available statistical methods which attempt to account for the impact of non-adherence to interventions were used infrequently. Therefore, firm inferences about their influence on non-inferiority conclusions could not be drawn. Since intention-to-treat and per-protocol analyses do not guarantee unbiased conclusions regarding non-inferiority, the methods identified should be considered for use in sensitivity analyses.

Study registration: Prospero CRD42020177458.

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ARTICLE SUMMARY

Strengths and limitations of this study

- This is the first systematic review (protocol published on PROSPERO) to identify statistical methods that attempt to account for the impact of non-adherence to interventions in randomised non-inferiority trials, quantify the use of such methods in these studies, and examine their impact on non-inferiority conclusions.
- A description and critique of the statistical methods identified is provided, along with their target estimands.
- Publications from any year, journal, or disease area/patient population were searched and a large number of papers (3,235) reviewed independently by two authors.
- Only one author extracted the data from the eligible papers.
- Whilst statistical analysis plans were requested for all eligible trials, these could not be obtained for all of the trials included.

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INTRODUCTION

Non-inferiority trials, which assess whether a new intervention is not worse than a proven comparator by more than a clinically acceptable amount, are becoming increasingly common.[1-3] They are principally used when it is hoped that the new intervention may convey some advantage other than better efficacy (its effect under ideal conditions), such as improved safety, tolerability, convenience, or reduced cost.[4, 5]

One of the challenges in these studies, and the focus of this review, is how participants not receiving their randomly assigned intervention according to the trial protocol (termed non-adherence or non-compliance) should be handled in the statistical analysis.[6] Such non-adherence is common in trials and has been associated with poorer health outcomes.[7-9] It can bias estimates of efficacy in either direction and so obtaining an accurate and reliable measure of adherence and accounting for any non-adherence in the statistical analysis of these studies is essential.[10, 11] The terms adherence and compliance are often used interchangeably, though adherence is preferred here since it is felt to better reflect the partnership between the healthcare provider and participant.

A simple approach to handling non-adherence is to define and analyse different analysis sets based on participants' observed levels of adherence, with consistent results providing greater confidence in the trial conclusions.[1] In the setting of non-inferiority trials, the intention-to-treat (ITT) and per-protocol (PP) populations have been advocated and are commonly used.[4, 12, 13] However, agreement between the ITT and PP results of these trials does not guarantee that conclusions regarding non-inferiority are free from bias caused by differential, or non-random, non-adherence (where the factors leading to non-adherence are associated with outcomes).[14-16]

Standard ITT analyses typically include all participants in their randomised groups irrespective of the intervention actually received.[17] Thus, they reflect the effect of assigning individuals to interventions

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in clinical practice where not everyone is fully adherent (also known as the 'effectiveness' of an intervention). This approach preserves the balance in known and unknown prognostic factors afforded by randomisation and so any difference in outcomes between study arms can be attributed solely to the experimental intervention.[18] However, in the presence of non-adherence, ITT analyses may yield biased estimates of efficacy (also known as the 'causal effect' of an intervention).[19] In non-inferiority trials, where efficacy and effectiveness may be considered equally important, this can increase the probability of falsely claiming non-inferiority and therefore accepting a worse intervention.[11]

Modified intention-to-treat (mITT) analyses are commonly used to address some of the limitations with standard ITT methods.[20] This approach allows some randomised participants, such as those who never receive any of the allocated intervention or who are identified as ineligible after randomisation, to be excluded in a justified way.[18] However, there is substantial variability in how this population is defined and bias may be introduced by subjectively excluding individuals from analysis.[18, 20] In addition, mITT analyses are not normally used to account for the impact of non-adherence.

PP analyses estimate the efficacy of interventions typically by excluding or censoring individuals with major protocol violations, including those who are non-adherent to their allocated intervention.[1, 6, 17] Excluding participants in this way can lead to selection bias because non-adherent individuals generally differ from those who are fully adherent with respect to prognostic factors.[21, 22] Furthermore, using a PP analysis to address differential non-adherence is likely to reduce the protection provided by randomisation, so that trial arms are not fully comparable; this potentially biases the study results in either direction.[11] In other words, any difference in outcomes between trial arms may no longer be due to the experimental intervention only. To obtain valid results from a PP analysis, we need to recover the protection due to randomisation, typically through a statistical method that (given certain assumptions) correctly adjusts for factors associated with both adherence and outcome (confounders).[21]

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Other statistical techniques that attempt to account for the impact of non-adherence and thus estimate the causal effects of experimental interventions exist. These range from simple approaches, such as including observed adherence as a covariate within a regression model, which like PP analyses is susceptible to selection bias, to more sophisticated techniques, such as instrumental variable (IV) methods and inverse-probability weighting, which allow for non-adherence while attempting to maintain the balance produced by randomisation.[23, 24] Several of these methods attempt to estimate the complier average causal effect (CACE), which is the causal effect of an intervention for individuals who would always be fully adherent regardless of assignment (known as compliers).[25] In other words, it is a comparison of the average outcome among those who are fully adherent in the experimental arm with the average outcome among the comparable group in the control arm who would fully adhere to the experimental intervention, if offered.

It is unclear which of the alternative methods have been applied in the setting of non-inferiority trials, to what extent, and with what results. Therefore, this systematic review aimed to identify statistical methods that adjust for the impact of non-adherence to interventions (thereby estimating the causal effects of experimental interventions) in randomised non-inferiority trials, quantify the use of such methods in these studies, and examine their impact on non-inferiority conclusions.

METHODS

The Ovid MEDLINE® database was searched for terms related to adherence, non-inferiority trials, and statistical methods for handling non-adherence in the titles, abstracts and key words of papers published up to 31st December 2020 (full search strategy provided in the appendix). Eligibility based on identifying appropriate statistical methods was assessed using a three-stage process. First, two authors independently reviewed the title and abstract of each paper. Those where the comparison

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was not randomised, the primary analysis was not for non-inferiority, or the analysis assessed cost-effectiveness were excluded (cost-effectiveness analyses were not of interest because the focus of this review was on estimating the efficacy of interventions). Papers not published in English were also excluded. Even if the full text was unavailable, the abstract was reviewed against the eligibility criteria to ensure that key papers were not excluded. Next, an automated search of the full texts was performed in order to identify those containing the terms 'compliance', 'adherence' or 'complier'. Finally, full-text reviews of the remaining papers were performed independently by two authors to identify, 1) randomised trials with a primary analysis for non-inferiority that applied (or planned to apply, for protocol papers) statistical methods to account for the impact of non-adherence to interventions, and 2) methodology papers that described such statistical methods and included a non-inferiority trial application. Any discrepancies between reviewer pairs were discussed with a third author in order to reach a consensus. In addition, statisticians within the field were consulted in order to identify key publications, and the reference lists and citations of eligible papers searched for relevant analyses (performed by one author (MD)). Meta-analyses and systematic reviews identified were also searched for eligible non-inferiority trials. Where a trial's published protocol and results paper were both eligible and reported the same statistical method of interest, the protocol paper was excluded to avoid double counting. Statistical analysis plans were requested for all eligible trials.

A standardised electronic form was used to extract the relevant information from each paper considered eligible. This included details of the trial characteristics (journal, year of publication, disease area or patient population, unit of randomisation, type of experimental intervention, type of primary outcome, statistical power, and non-inferiority margin), non-adherence to the interventions (definitions and estimated levels of non-adherence), the statistical method attempting to account for non-adherence (name of the method, estimand, estimate of effect and confidence interval, conclusion regarding non-inferiority, and any advantages/disadvantages of the method stated), and any other analyses applied to the same outcome (analysis population, estimand, estimate of effect and

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confidence interval, and conclusion regarding non-inferiority). Data extraction was performed by one author (MD). The primary outcome was the statistical method applied (or planned to be applied) in order to account for non-adherence to the interventions. Other outcomes were the impact of applying these methods on the trial conclusions (compared with other analyses applied to the same outcome, where available) and the advantages and disadvantages of the methods where stated by the authors. The impact of applying the methods of interest was assessed using trial results papers only.

This systematic review was registered with Prospero (registration CRD42020177458) and conducted in accordance with the PRISMA statement.[26] All analyses were conducted using Stata version 15.1.

Patient and public involvement

Patients or the public were not involved in the design, conduct, reporting, or dissemination plans of our research.

RESULTS

After removing duplicate publications, our search identified 3,235 papers. Of these, 934 were excluded following review of the titles and abstracts, 790 did not contain any key words in the full texts, and 1,489 were excluded after full-text review, leaving 22 papers whose citations and reference lists contained a further five papers meeting eligibility criteria. After removing publications of the same trial reporting identical statistical methods of interest, 24 papers remained (Figure 1).

The 24 publications, which consisted of four protocols, 13 results papers, and seven methodology papers, reported relevant methods on 26 occasions (two methodology papers both contained two relevant analyses). Four of the analyses included in methodology papers were re-analyses of non-inferiority trials, one included a simulation study based on a non-inferiority trial, and four included simulation studies not based on real trials. Fifteen of the 24 papers included (63%) were published

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within the last five years and the most common type of experimental intervention studied was drug interventions (35%) (Table 1).

Non-adherence to interventions

Non-adherence to the randomly assigned interventions was defined in the methods, statistical analysis plan, or results section of most analyses (n=19, 73%). Fifteen (79%) used a binary definition of adherence, whereas three (16%) used a continuous measure (one was unclear). Of the 19 analyses that defined non-adherence to the interventions, 13 reported estimates of non-adherence (the remaining six were protocols or simulation studies). More than half reported estimates of non-adherence that were no more than 10%, though the range was wide (1.7% to 51.3%) (Table 2). For reasons that were not reported, two papers provided data on non-adherence to the interventions in only one arm of the trial.

Statistical methods for handling non-adherence to interventions

In total, 11 different statistical methods that attempt to account for non-adherence to interventions were identified (Table 3). The most common were IV approaches (n=9, 35%), including observed adherence as a covariate within a regression model (n=3, 12%), and modelling adherence as a time-varying covariate in a time-to-event analysis (n=3, 12%). Other methods included rank preserving structural failure time models and G-estimation (n=2, 8%), inverse-probability-of-treatment weighting (n=2, 8%) and the tipping point approach (n=2, 8%, both in the same methodology paper). The other five techniques identified were all reported once. Further details of the methods reported more than once are provided in Table 4. The techniques identified in the 17 protocols and results papers were more commonly specified as sensitivity analyses (n=13, 76%) than primary analyses (n=3, 18%) (one was unclear).

Impact of the statistical methods on non-inferiority conclusions

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Twelve of the 13 results papers (92%) also included an alternative analysis of the same outcome (Table A1 in the appendix). All 12 performed an ITT or mITT analysis. In addition, some reported results from PP (n=6, 50%) or as-treated (AT; n=2, 17%) analyses. Non-inferiority conclusions from the alternate analyses were in agreement with those from the methods of interest on six occasions and could not be compared on six occasions (due to different measures of effect or the results not being provided in full). Five of the six analyses where the different methods were in agreement concluded non-inferiority of the experimental intervention versus the comparator. The remaining trial provided mixed findings regarding non-inferiority across the two different countries included, though the interpretation of this study appeared inconsistent with its design (a confidence interval approach to determining non-inferiority was stated in the methods but not utilised).

DISCUSSION

To our knowledge, this is the first systematic review undertaken to both identify statistical methods that adjust for the impact of non-adherence to interventions in randomised non-inferiority trials and also identify the frequency and consequences of their use. We found that few papers reported such methods (less than 2% of those reaching full-text review). This may be partly due to unfamiliarity with such techniques amongst trialists and statisticians as a result of the long lead time for statistical methodology to make its way into routine practice. The most common techniques identified were IV approaches, including observed adherence as a covariate within a regression model, and modelling adherence as a time-varying covariate in a time-to-event analysis. Overall, the number of trials implementing relevant statistical methods was too small to draw firm inferences about their impacts on non-inferiority conclusions. In six analyses where the results from methods of interest could be compared directly with those from an alternative analysis, conclusions regarding non-inferiority were consistent across the different approaches.

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Almost half of the methods identified focus on estimating CACE (also known as the local average treatment effect (LATE)). This is the average effect of the experimental intervention within the subpopulation of compliers.[25] We argue that this is the natural estimation focus when attempting to account for non-adherence to interventions in the context of non-inferiority trials. This is because we want to be confident that there is non-inferiority among those who would comply with either intervention. By contrast, including participants who would not fully adhere to both interventions may bias estimation towards non-inferiority (in a similar way that, in the context of non-inferiority, ITT analyses may be biased towards non-inferiority under non-adherence). For similar reasons, we believe that the CACE is preferable to the population average treatment effect (ATE). Lastly, we note that when adjusting for observed adherence within a regression model or modelling adherence as a time-varying covariate in a time-to-event analysis, the target estimand is unclear.

The infrequent use of statistical methods for handling non-adherence seen in the current review has also been observed more generally in randomised controlled trials (RCTs). A review of one hundred RCTs randomly selected from those published in four high-impact journals during 2008 found only one that attempted to account for non-adherence to interventions using a causal inference framework (in which inverse-probability-of-censoring weighting was applied).[6] More recently, Mostazir et al. conducted a review of statistical approaches for handling non-adherence to interventions in RCTs published between 1991 and 2015 which identified 88 analyses incorporating nine different methods.[27] IV methods were among the most common and accounted for almost one in four applications of suitable techniques. However, some of the other methods identified (including CACE analyses using maximum-likelihood estimation and adjusted treatment received models) were not captured in the current review focussing on non-inferiority trials. Similarly, we did not identify all twelve approaches included in a recent review of methodological papers containing statistical techniques for handling non-adherence to interventions in the context of time-to-event outcomes.[28] This suggests that other relevant methods are available but either they are not suitable for comparing

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active interventions, as is often required in non-inferiority trials, or they may not have been applied within these studies. The three aforementioned reviews did not focus specifically on non-inferiority trials.

It is perhaps not surprising that IV approaches were the most common method identified in the current review, given that their assumptions are well suited to many double-blind trials, they can be applied across a range of trial designs, and they are relatively simple to implement in standard statistical software.[29] Their main assumptions are: 1) randomisation affects the outcome only through its influence on the intervention received (the *exclusion restriction*), 2) randomisation does not share common causes with the outcome (the *exchangeability* assumption), 3) randomisation causes some participants to receive their assigned intervention (the *relevance* assumption), and, in order to estimate CACE, 4) there are no participants who would always receive the opposite of their random allocation (the *monotonicity* assumption).[23, 30] In individually randomised trials, the exclusion restriction and monotonicity assumptions are typically satisfied by effective double-blinding and use of objective outcomes, and the exchangeability assumption is usually valid since randomisation is expected to produce trial arms that are balanced with respect to prognostic factors. Under these assumptions, IV methods enable us to estimate CACE even in the presence of unmeasured confounding.[31] Therefore, these approaches do not require measurement of all confounders, though their inclusion in the analysis can improve precision.[32] A lack of precision and statistical power is often a challenge with IV techniques and methods used to adjust for non-adherence more generally.[5, 29, 33, 34]

Another limitation of IV methods is that when interventions are administered at multiple timepoints, standard approaches are susceptible to time-varying confounding and selection bias.[21] These biases occur when previous values of a covariate predict the current intervention received and the current value of the covariate predicts outcome.[35] If the time-varying confounders are themselves affected by

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previous intervention received, so-called G-methods, such as inverse-probability weighting or G-estimation, are required to allow for the feedback loop occurring between the intervention received and confounders over time.[21, 24, 36] G-methods were seldom reported in the current review, perhaps because they can be more complex to implement than alternative approaches and also rely on assumptions which may be vulnerable to violations. For instance, inverse-probability weighting relies on the assumption of no unmeasured confounding and rank preserving structural failure time models assume a constant effect of the experimental intervention, irrespective of when it is received.[37]

In order to estimate the effect of the experimental intervention in the absence of (full) protection by randomisation, additional assumptions must be made. Most of these are, by their nature, inherently untestable. Each of the statistical methods identified in the current review make slightly different assumptions in order to estimate the effects of interventions under full adherence, and hence each has a different method of estimation; both assumptions and estimation methods have associated advantages and disadvantages. Crucially, all of the methods require reliable information regarding adherence to the randomly assigned interventions, which is often challenging to measure, particularly for long-term therapies.[38]

Despite these limitations, it is our view that the methods identified have an important role in non-inferiority trials with non-adherence to interventions and should be applied as sensitivity analyses alongside other techniques (such as ITT, PP and AT analyses). Given that agreement between ITT and PP analyses cannot guarantee unbiased conclusions in these studies, those with non-trivial non-adherence should assess the sensitivity of trial results to different assumptions in order to guard against falsely claiming non-inferiority and accepting a worse intervention. Careful consideration needs to be given to the assumptions that are most plausible given the trial context and planned design, before selecting the appropriate statistical method to adjust for potential non-adherence. Relevant data needed to implement the chosen technique should then be collected as fully as possible.

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Clearly, the best approach to reducing the potential biases introduced by non-adherence to the interventions is to design trials that minimise such non-adherence. Future work should compare the performance of the methods identified under different non-adherence scenarios in non-inferiority trials in order to facilitate understanding of when they might be applied appropriately.

Strengths and limitations

This is the first systematic review (protocol published on PROSPERO) to identify statistical methods that attempt to account for the impact of non-adherence to interventions in randomised non-inferiority trials, quantify the use of such methods in these studies, and examine their impact on non-inferiority conclusions. The review included publications from any year, journal or disease area and involved two authors agreeing the eligibility of each paper identified in the search. However, it has some limitations. First, the search for eligible papers had to be restricted to those containing terms related to adherence and statistical methods for handling non-adherence in the titles, abstracts and key words. Publications applying suitable methods in sensitivity analyses may not have referred to the techniques within these fields and, if so, would not have been captured by the search. A wide range of search terms were used to try and mitigate this problem and therefore a large number of papers reviewed. Second, only one author performed the data extraction from eligible publications, though other authors were consulted where necessary. Finally, whilst statistical analysis plans were requested, these could not be obtained for eight of the trials included. For these studies, we cannot be sure that the details provided in the publications reviewed are accurate accounts of the planned analyses.

CONCLUSION

In non-inferiority trials with non-adherence to interventions, ITT and PP analyses are often performed but may result in biased estimates of efficacy and therefore agreement between these approaches does not guarantee that conclusions regarding non-inferiority are unbiased. Statistical methods that

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attempt to account for the impact of non-adherence and thereby estimate the causal effects of interventions are available, but their use in non-inferiority trials remains extremely infrequent. It is our view that the methods identified should be applied more widely within sensitivity analyses of non-inferiority trials. In particular, those with non-trivial non-adherence should assess the sensitivity of trial results to different assumptions in order to guard against falsely claiming non-inferiority and accepting a worse intervention.

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Author contributions

MD devised the review, reviewed the eligibility of papers identified, carried out data extraction and analysis, and wrote the initial manuscript. KF, JC, JAT and DE commented on the review protocol, reviewed the eligibility of papers identified, and assisted with writing the final manuscript.

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Competing interests

None declared.

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Data sharing statement

No additional data are available.

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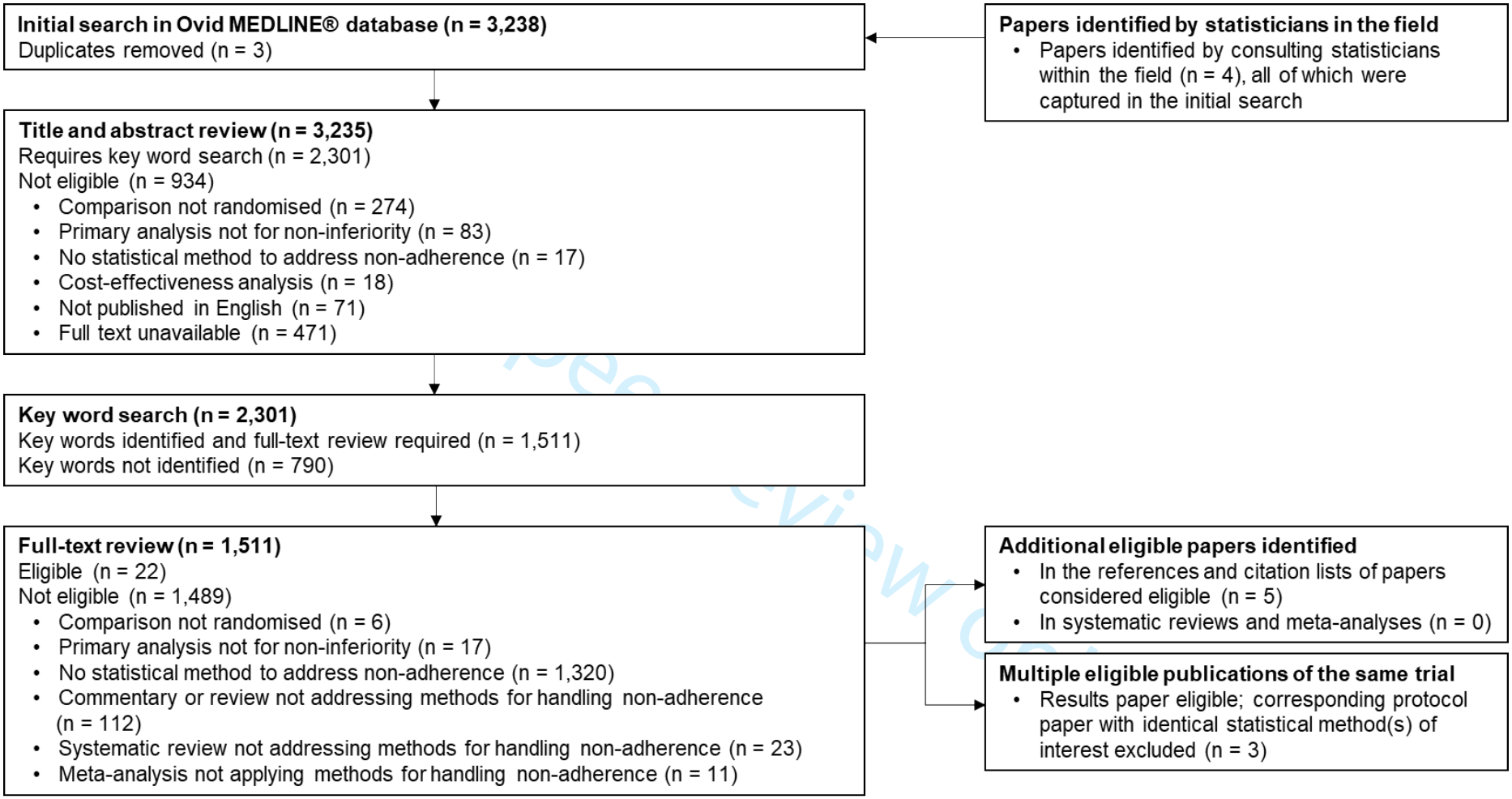


Figure 1. Flow chart showing the eligibility of papers reviewed

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Table 1. Characteristics of eligible analyses (n=26)

Characteristics	n (%)
Type of publication (n=24)	
Results	13 (54)
Methodology	7 (29)
Protocol	4 (17)
Journal (n=24)	
The New England Journal of Medicine	4 (17)
Statistics in Medicine	3 (13)
BJOG	2 (8)
Journal of Biopharmaceutical Statistics	2 (8)
Trials	2 (8)
Addiction	1 (4)
American Journal of Kidney Disease	1 (4)
BMC Psychiatry	1 (4)
BMJ Open	1 (4)
Clinical Gastroenterology and Hepatology	1 (4)
Clinical Pediatrics	1 (4)
Food and Nutrition Bulletin	1 (4)
The Journal of the American Medical Association (JAMA)	1 (4)
JAMA Network Open	1 (4)
PLOS Medicine	1 (4)
Wellcome Open Research	1 (4)
Year of publication (n=24)	
2006 – 2010	5 (21)
2011 – 2015	4 (17)
2016 – 2020	15 (63)
Disease area or patient population	
Mental health	4 (15)
Appendicitis	2 (8)
Cancer	2 (8)
Respiratory infection/disease	2 (8)
Ulcerative colitis	2 (8)
Anaemia	1 (4)
Ear infection	1 (4)
General surgery patients	1 (4)
Heart disease	1 (4)
Human immunodeficiency virus (HIV)	1 (4)
Individuals receiving life-sustaining therapies	1 (4)
Renal disease	1 (4)
Smoking cessation	1 (4)
Throat infection	1 (4)
Urinary incontinence	1 (4)
Simulation study*	4 (15)
Unit of randomisation	
Individual	20 (77)
Cluster	3 (12)
Simulation study	3 (12)

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Type of experimental intervention	
Drug	9 (35)
Method of treatment delivery	3 (12)
Additional patient examination	2 (8)
Nutritional	2 (8)
Surgical	1 (4)
Simulation study*	4 (15)
Other	5 (19)
Type of outcome	
Binary	13 (50)
Continuous	8 (31)
Time-to-event	4 (15)
Count	1 (4)
Composite outcome	3 (12)
Statistical power	
80%	13 (50)
90%	6 (23)
At least 80%	3 (12)
Other	1 (4)
Not stated	3 (12)

* not based on real non-inferiority trials; as reported in the statistical methods.

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Table 2. Estimates of non-adherence to interventions reported in methodology and results papers, combined across trial arms unless reported (n=13)

Estimate of non-adherence	Binary measure of adherence (n=11)	Continuous* measure of adherence (n=2)	Binary or continuous* measure of adherence (n=13)
≤5%	4 (36)	0 (0)	4 (31)
6 – 10%	4 (36)	0 (0)	4 (31)
11 – 25%	0 (0)	1 (50)	1 (8)
26 – 50%	2 (18)†	1 (50)	3 (23)
>50%	1 (9)†	0 (0)	1 (8)

Data presented as n (%). * Mean level of non-adherence; † Two papers provided an estimate of non-adherence in only one arm of the trial.

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Table 3. Statistical methods that were identified as attempting to account for non-adherence to interventions

Method (estimand)	Brief description	n (%)	Advantages*	Disadvantages*
IV approaches (CACE)	CACE estimated using the conventional IV estimator, the Generalized Method of Moments IV estimator, or two-stage least squares (2SLS) regression	9 (35)	1) Straightforward to compute[29] 2) Preserves the balance in patient characteristics from randomization[29] 3) Although the validity of the IV estimator depends on several key assumptions, these assumptions are likely to hold in most double-blinded studies[29] 4) Correctly adjusts for missing outcome data, assuming that these are missing at random (MAR)[39] 5) Can account for unknown confounders[11] 6) Recent methods using doubly robust procedures have been developed to boost power when using IV estimation[11]	1) Accurate data on compliance behaviour must also be available[29] 2) The IV method… does increase the sample size requirements of the study as the expected proportion of noncompliers increases[29] 3) Requires the "exclusion restriction" to be fulfilled (i.e. treatment allocation only influences the outcome through the treatment and not through any other pathways). This assumption is unverifiable and we are only likely to be confident that it holds in a double blinded study.[11]
Adjustment for observed adherence (estimand unclear)	Observed adherence included as a covariate within a regression model	3 (12)	None stated	None stated
Adherence modelled as a time-varying covariate in a time-to-event analysis (estimand unclear)	Attributes the time at risk between observations and the outcomes occurring during the same period to the concurrent value of adherence	3 (12)	None stated	1) Results… (from the) time-dependent analysis could be the consequence of a selection bias[40]
Tipping point approach (estimates the probability of reversing the trial conclusions under a range of assumptions about the outcome	Outcome data following non-adherence is treated as missing. Assesses how sensitive the trial results are to the values of these missing outcomes.	2 (8)†	1) A model or mechanism for the missing outcomes does not have to be assumed[41] 2) All randomised individuals are included in the analysis[41]	1) Require(s) assumptions that are often difficult or not possible to verify[41]

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Method (estimand)	Brief description	n (%)	Advantages*	Disadvantages*
data following non-adherence)				
Rank preserving structural failure time model and G-estimation (ATE)	Untreated survival times (those that would occur if no EXP were received) are assumed to be a function of both time on/off EXP and the effect of EXP compared with CON. The value for the effect of EXP that results in equal untreated survival times in the randomised groups is identified (via G-estimation) and used to calculate adjusted survival times that would have been observed had no switching occurred.	2 (8)	1) Takes compliance history into account[42] 2) Maintains the original randomized group[43] 3) (G-estimation) can be performed for other types of outcomes, such as continuous, binary, or count responses, using structural nested mean models[43]	1) Can only be used under a specific set of assumptions[42] 2) The rank preserving structural failure time model… incorporates a strong non-interaction assumption with respect to the treatment effect[43]
Inverse-probability-of-treatment weighting (ATE)	A pseudo-population is created by re-weighting participants' outcomes according to the probability of adherence at each visit, given previous values of the intervention received and confounders. The causal effect of EXP is estimated by performing an unadjusted analysis in the pseudo-population, which is equivalent to a weighted analysis in the original cohort.	2 (8)	1) Ensure(s) that the reweighted arms are similar and comparable[11] 2) Sensitivity analysis methods are available to address unobserved confounding and covariate measurement errors[11]	1) It eliminates bias if all confounders can be appropriately adjusted for, but in general this will not be possible[11]
Structural mean models (CACE)	Baseline variables that predict adherence differentially in each arm of the trial and are also conditionally independent of outcome are identified. Enables the estimation of two distinct causal parameters, from which a contrast can be made.	1 (4)	1) Straightforward to implement using standard statistical software[34]	1) This paper highlights the increase in variance experienced when fitting these models, something that can only be reduced when the models include strong predictors of adherence and outcome[34] 2) Extension of the approach to handle nonlinear outcome variables is also required[34]

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Method (estimand)	Brief description	n (%)	Advantages*	Disadvantages*
CACE analysis using propensity score approach (CACE)	A propensity score is developed in the EXP group in order to predict the probability that those in the CON group would have been fully adherent if assigned to EXP. CON group outcomes are re-weighted using these probabilities and compared with the outcomes of those who were fully adherent in the EXP group.	1 (4)	None stated	None stated
CACE analysis using a mixture modelling approach (CACE)	A mixture model is used to identify those in the CON group that are likely to have been fully adherent had they been assigned to EXP. Outcomes in this subgroup are compared with outcomes of those who were fully adherent in the EXP group.	1 (4)	None stated	None stated
Test statistic based on the OR in CRTs (estimand unclear)	A test statistic for assessing non-inferiority based on the OR in CRTs under the Dirichlet multinomial model	1 (4)	None stated	None stated
CACE analysis (CACE)	Exact method not stated	1 (4)	None stated	None stated

* As stated by the authors; † both applications of the tipping point approach were reported in the same methodology paper; IV = instrumental variable; CACE = complier average causal effect; ATE = average treatment effect in the population; EXP = experimental intervention; CON = control; OR = odds ratio; CRT = cluster randomised trial.

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Table 4. Details of the statistical methods reported on more than one occasion

Method	Description	Key assumptions	Advantages	Disadvantages
IV approaches[23, 30]	With two randomised groups (*Z*; *Z*=0 for those in the CON group, *Z*=1 for those in the EXP group), a binary measure of the intervention received (*X*; *X*=1 when EXP received, *X*=0 when CON received), and a continuous outcome (*Y*), the conventional IV estimator is: $IV = \frac{E[Y \mid Z=1] - E[Y \mid Z=0]}{\Pr(X=1 \mid Z=1) - \Pr(X=1 \mid Z=0)}$ Alternatively, 2SLS regression can be used in the presence of confounding by baseline covariates (*C*). First, *X* is regressed on *Z* and *C.* Then, *Y* is regressed on both the predicted value of *X* obtained from the first regression and *C*. The coefficient for the effect of *X* on *Y* in this second model provides an estimate of CACE.	The instrument (randomisation): 1) Affects the outcome only through the intervention received (the *exclusion restriction*) 2) Does not share common causes with the outcome (the *exchangeability* assumption) 3) Causes some participants to receive their assigned intervention (the *relevance* assumption) Additional assumption required to estimate CACE: 4) There are no participants who would always receive the opposite of their random allocation (the *monotonicity* assumption)	1) Preserves randomised comparison[29] 2) Assumptions well suited to double-blinded trials (typically assumptions 1 and 4 are satisfied by effective double-blinding and use of objective outcomes, and assumption 2 valid due to randomisation)[31] 3) Under assumptions 1-4, able to estimate CACE even when unmeasured confounding is present[31] 4) Inclusion of confounders in 2SLS regression can improve precision[32] 5) Can be extended to allow for partial adherence, binary or time-to-event outcomes, and clustering, though additional assumptions may be required[44-48]	1) Requires untestable assumptions (1 and 4) that may be violated. When adherence is binary, the exclusion restriction implies no effect of EXP in those who are non-adherent. 2) Only appropriate when crossovers occur and cannot be used when non-trial interventions are received[5] 3) The sample size required to maintain statistical power to detect non-inferiority increases as the quantity of non-adherence increases[5] 4) Simple approaches described involve a single measure of the intervention received (e.g. ≥80% of sessions attended versus <80%, or the proportion of sessions attended) and are susceptible to time-varying confounding (where predictors of both adherence and outcome vary over time)
Adjustment for observed adherence	Observed adherence at a fixed timepoint (e.g. whether surgery was performed) or multiple timepoints (e.g. whether medication was taken adequately between follow-up visits) included as a covariate within a regression model	1) Individuals in the EXP and CON groups with the same level of observed adherence are comparable 2) Within each trial arm, the effect of adherence on the outcome is the same	1) Relatively straightforward to implement	1) Susceptible to selection bias and should be avoided. Different factors may lead to non-adherence in the two groups, meaning those in the EXP group with an observed level of adherence may differ from those in the CON group with the same level of

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Method	Description	Key assumptions	Advantages	Disadvantages
		3) The functional form of adherence is correctly specified 4) Other model assumptions are not violated		adherence. Also, some of those in the CON group that were non-adherent may have been fully adherent if assigned to the EXP group (and vice versa).[32] 2) Does not account for time-varying confounding
Adherence modelled as a time-varying covariate in a time-to-event analysis[49]	An extension of the Cox PH model that allows the intervention received to vary over time. The model takes the form: $$\lambda_i(t) = \lambda_0(t)\exp(\beta X_i(t))$$ where $\lambda_i(t)$ is the hazard function at time t, $\lambda_0(t)$ is the baseline hazard function, and $X_i(t)$ takes the value 0 while a participant receives CON and 1 while they receive EXP. β is the log hazard ratio for the effect of receiving EXP versus CON.	1) Only the current value of the intervention (at time t) affects the hazard 2) The effect of receiving EXP is the same for all participants regardless of when it is received 3) Other assumptions of the Cox PH model (including uninformative censoring) are not violated	1) Allows for time-varying confounding that is not influenced by previous intervention received 2) Can be extended to allow for more flexible measures of the intervention received, such as cumulative exposure to the intervention 3) Non-trial interventions may be incorporated	1) Assumptions 1 and 2 are difficult to verify and may be violated 2) Susceptible to selection bias if switching is related to prognostic factors[49] 3) Does not account for time-varying confounding that is influenced by previous intervention received
Tipping point approach[41]	Outcome data following non-adherence is treated as missing. A range of assumptions about these outcomes are explored to assess how sensitive the trial results are to the missing values.	Assumptions made about the values of missing outcomes following non-adherence e.g. all missing outcomes are, 1) failures, 2) successes in the CON group and failures in the EXP group (worst case scenario), or 3) failures in the CON group and successes in the EXP group (best case scenario)	1) All randomised individuals are included in the analysis[41] 2) A model or mechanism for the missing outcomes does not have to be assumed[41]	1) While a range of assumptions about the missing values can be explored, these assumptions are often difficult or not possible to verify[41]
Rank preserving structural failure time model and G-	Let T_i denote the observed survival time for the ith participant and U_i their survival time that would have been observed if they received no EXP. T_i is assumed to be a function of time on (	1) If no participants received any EXP, the average survival time in the two groups would be equal (due to randomisation)	1) Preserves randomised comparison[43] 2) Takes adherence history into account[42]	1) Assumption 2 and 3 are difficult to verify and may be violated[49] 2) Additional assumptions required when the CON group receive an active intervention[50]

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Method	Description	Key assumptions	Advantages	Disadvantages
estimation[36, 49, 50]	T_{ON_i}) and time off (T_{OFF_i}) EXP and T_i related to U_i by the causal model: $$U_i = T_{OFF_i} + e^{-\psi_0} T_{ON_i}$$ e^{ψ_0} is the amount by which expected survival times are increased by EXP (the *acceleration factor*). Due to randomisation, U_i is assumed to be independent of trial arm. Untreated event times are predicted for all participants and the value of ψ_0 that results in equal untreated survival times in the randomised groups identified (using G-estimation). This value of ψ_0 is used to calculate adjusted survival times that would have been observed had switching not occurred.	2) The effect of receiving EXP is the same for all participants regardless of when it is received 3) If participant *i* experiences the event of interest before participant *j* when both are treated, then participant *i* would also experience the event before participant *j* when both are untreated (*rank preserving*)	3) Allows for time-varying confounding, including when these confounders are affected by previous intervention received 4) Does not require information on potential confounders (only randomised group, observed event times, and intervention history)[51]	3) Only appropriate when crossovers occur and cannot be used when non-trial interventions are received [51] 4) G-estimation may not work well if the number of participants or events is small[50] 5) Computationally intensive[52]
Inverse-probability-of-treatment weighting[24, 52-54]	Confounding is accounted for by re-weighting participants' outcomes. Typically, logistic regression is used to predict probabilities of adherence at each visit given previous values of the intervention received and confounders. The inverse of these probabilities (the weights) are used to create a pseudo-population in which time-varying confounders are not associated with the intervention. The causal effect of EXP is estimated by performing an unadjusted analysis in the pseudo-population, which is equivalent to a weighted analysis in the original cohort.	1) The absence of unmeasured confounding (*exchangeability*) 2) Participants have a non-zero probability of receiving each intervention (*positivity*) 3) The observed outcome equals the counterfactual outcome of the intervention actually received (*consistency*) 4) No misspecification of the model used to estimate the weights	1) Preserves randomised comparison[11] 2) Takes adherence history into account 3) Allows for time-varying confounding, including when these confounders are affected by previous intervention received 4) Can be extended to allow for non-trial interventions, but may result in large weights and estimates very sensitive to model specification[53]	1) Eliminates bias if all confounders can be appropriately adjusted for, but in general this will not be possible[11] 2) Cannot be used if covariates perfectly predict adherence[52] 3) Unstable in the presence of extreme weights[52]

CACE = complier average causal effect; IV = instrumental variable; CON = control; EXP = experimental intervention; 2SLS = two-stage least squares; PH = proportional hazards.

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REFERENCES

1. Piaggio G, Elbourne DR, Altman DG, Pocock SJ, Evans SJ. Reporting of noninferiority and equivalence randomized trials: an extension of the CONSORT statement. Jama. 2006 Mar 8;295(10):1152-60
2. Suda KJ, Hurley AM, McKibbin T, Motl Moroney SE. Publication of noninferiority clinical trials: changes over a 20-year interval. Pharmacotherapy. 2011 Sep;31(9):833-9
3. Bikdeli B, Welsh JW, Akram Y, Punnanithinont N, Lee I, Desai NR, et al. Noninferiority Designed Cardiovascular Trials in Highest-Impact Journals. Circulation. 2019 Jul 30;140(5):379-89
4. Piaggio G, Elbourne DR, Pocock SJ, Evans SJ, Altman DG. Reporting of noninferiority and equivalence randomized trials: extension of the CONSORT 2010 statement. Jama. 2012 Dec 26;308(24):2594-604
5. Mo Y, Lim C, Watson JA, White NJ, Cooper BS. Non-adherence in non-inferiority trials: pitfalls and recommendations. Bmj. 2020 Jul 1;370:m2215
6. Dodd S, White IR, Williamson P. Nonadherence to treatment protocol in published randomised controlled trials: a review. Trials. 2012 Jun 18;13:84
7. DiMatteo MR. Variations in patients' adherence to medical recommendations: a quantitative review of 50 years of research. Med Care. 2004 Mar;42(3):200-9
8. DiMatteo MR, Giordani PJ, Lepper HS, Croghan TW. Patient adherence and medical treatment outcomes: a meta-analysis. Med Care. 2002 Sep;40(9):794-811
9. Simpson SH, Eurich DT, Majumdar SR, Padwal RS, Tsuyuki RT, Varney J, et al. A meta-analysis of the association between adherence to drug therapy and mortality. Bmj. 2006 Jul 1;333(7557):15
10. Ye C, Beyene J, Browne G, Thabane L. Estimating treatment effects in randomised controlled trials with non-compliance: a simulation study. BMJ Open. 2014 Jun 17;4(6):e005362
11. Mo Y, Lim C, Mukaka M, Cooper BS. Statistical considerations in the design and analysis of non-inferiority trials with binary endpoints in the presence of non-adherence: a simulation study. Wellcome Open Res. 2019;4:207
12. Committee for Proprietary Medicinal Products (CHMP). Points to consider on switching between superiority and non-inferiority. Br J Clin Pharmacol. 2001 Sep;52(3):223-8
13. Rehal S, Morris TP, Fielding K, Carpenter JR, Phillips PP. Non-inferiority trials: are they inferior? A systematic review of reporting in major medical journals. BMJ Open. 2016 Oct 7;6(10):e012594
14. Matilde Sanchez M, Chen X. Choosing the analysis population in non-inferiority studies: per protocol or intent-to-treat. Stat Med. 2006 Apr 15;25(7):1169-81
15. Brittain E, Lin D. A comparison of intent-to-treat and per-protocol results in antibiotic non-inferiority trials. Stat Med. 2005 Jan 15;24(1):1-10
16. Clarke PS, Windmeijer F. Identification of causal effects on binary outcomes using structural mean models. Biostatistics. 2010 Oct;11(4):756-70
17. ICH Harmonised Tripartite Guideline. Statistical principles for clinical trials. International Conference on Harmonisation E9 Expert Working Group. Stat Med. 1999 Aug 15;18(15):1905-42
18. Gupta SK. Intention-to-treat concept: A review. Perspect Clin Res. 2011 Jul;2(3):109-12
19. Nagelkerke N, Fidler V, Bernsen R, Borgdorff M. Estimating treatment effects in randomized clinical trials in the presence of non-compliance. Stat Med. 2000 Jul 30;19(14):1849-64
20. Abraha I, Montedori A. Modified intention to treat reporting in randomised controlled trials: systematic review. BMJ. 2010;340:c2697
21. Hernán MA, Robins JM. Per-Protocol Analyses of Pragmatic Trials. N Engl J Med. 2017 Oct 5;377(14):1391-8

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

22. DeMets DL. Statistical issues in interpreting clinical trials. J Intern Med. 2004 May;255(5):529-37
23. Angrist JD, Imbens GW, Rubin DB. Identification of Causal Effects Using Instrumental Variables. Journal of the American Statistical Association. 1996;91(434):444-55
24. Robins JM, Hernán MA, Brumback B. Marginal structural models and causal inference in epidemiology. Epidemiology. 2000 Sep;11(5):550-60
25. Yau LHY, Little RJ. Inference for the Complier-Average Causal Effect From Longitudinal Data Subject to Noncompliance and Missing Data, With Application to a Job Training Assessment for the Unemployed. Journal of the American Statistical Association. 2001 2001/12/01;96(456):1232-44
26. Page MJ, Moher D, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. PRISMA 2020 explanation and elaboration: updated guidance and exemplars for reporting systematic reviews. BMJ. 2021;372:n160
27. Mostazir M, Taylor RS, Henley W, Watkins E. An overview of statistical methods for handling nonadherence to intervention protocol in randomized control trials: a methodological review. J Clin Epidemiol. 2019 Apr;108:121-31
28. Alshreef A, Latimer N, Tappenden P, Wong R, Hughes D, Fotheringham J, et al. Statistical Methods for Adjusting Estimates of Treatment Effectiveness for Patient Nonadherence in the Context of Time-to-Event Outcomes and Health Technology Assessment: A Systematic Review of Methodological Papers. Med Decis Making. 2019 Nov;39(8):910-25
29. Kim MY. Using the instrumental variables estimator to analyze noninferiority trials with noncompliance. J Biopharm Stat. 2010 Jul;20(4):745-58
30. Dunn G, Maracy M, Tomenson B. Estimating treatment effects from randomized clinical trials with noncompliance and loss to follow-up: the role of instrumental variable methods. Stat Methods Med Res. 2005 Aug;14(4):369-95
31. Lousdal ML. An introduction to instrumental variable assumptions, validation and estimation. Emerg Themes Epidemiol. 2018;15:1
32. Stuart EA, Perry DF, Le HN, Ialongo NS. Estimating intervention effects of prevention programs: accounting for noncompliance. Prev Sci. 2008 Dec;9(4):288-98
33. Davies NM, Smith GD, Windmeijer F, Martin RM. Issues in the reporting and conduct of instrumental variable studies: a systematic review. Epidemiology. 2013 May;24(3):363-9
34. Gillespie D, Farewell D, Barrett-Lee P, Casbard A, Hawthorne AB, Hurt C, et al. The use of randomisation-based efficacy estimators in non-inferiority trials. Trials. 2017 Mar 9;18(1):117
35. Daniel RM, Cousens SN, De Stavola BL, Kenward MG, Sterne JA. Methods for dealing with time-dependent confounding. Stat Med. 2013 Apr 30;32(9):1584-618
36. Hernán MA, Cole SR, Margolick J, Cohen M, Robins JM. Structural accelerated failure time models for survival analysis in studies with time-varying treatments. Pharmacoepidemiol Drug Saf. 2005 Jul;14(7):477-91
37. Latimer NR, White IR, Tilling K, Siebert U. Improved two-stage estimation to adjust for treatment switching in randomised trials: g-estimation to address time-dependent confounding. Stat Methods Med Res. 2020 Oct;29(10):2900-18
38. Lam WY, Fresco P. Medication Adherence Measures: An Overview. BioMed Research International. 2015 2015/10/11;2015:217047
39. Kitchener HC, Dunn G, Lawton V, Reid F, Nelson L, Smith AR. Laparoscopic versus open colposuspension--results of a prospective randomised controlled trial. Bjog. 2006 Sep;113(9):1007-13
40. Brunori G, Viola BF, Parrinello G, De Biase V, Como G, Franco V, et al. Efficacy and safety of a very-low-protein diet when postponing dialysis in the elderly: a prospective randomized multicenter controlled study. Am J Kidney Dis. 2007 May;49(5):569-80

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

41. Kim M, Wang C, Xue X. Assessing the influence of treatment nonadherence on noninferiority trials using the tipping point approach. Stat Med. 2019 Feb 20;38(4):650-9
42. Wu Y, Zhao L, Hou Y, Li K, Zhou X. Correcting for non-compliance in randomized non-inferiority trials with active and placebo control using structural models. Stat Med. 2015 Mar 15;34(6):950-65
43. Matsuyama Y. A comparison of the results of intent-to-treat, per-protocol, and g-estimation in the presence of non-random treatment changes in a time-to-event non-inferiority trial. Stat Med. 2010 Sep 10;29(20):2107-16
44. Angrist JD, Imbens GW. Two-Stage Least Squares Estimation of Average Causal Effects in Models with Variable Treatment Intensity. Journal of the American Statistical Association. 1995 1995/06/01;90(430):431-42
45. Nagelkerke N, Fidler V, Bernsen R, Borgdorff M. Estimating treatment effects in randomized clinical trials in the presence of non-compliance. Statistics in Medicine. 2000;19(14):1849-64
46. Palmer TM, Thompson JR, Tobin MD, Sheehan NA, Burton PR. Adjusting for bias and unmeasured confounding in Mendelian randomization studies with binary responses. International Journal of Epidemiology. 2008;37(5):1161-8
47. Tchetgen Tchetgen EJ, Walter S, Vansteelandt S, Martinussen T, Glymour M. Instrumental variable estimation in a survival context. Epidemiology. 2015 May;26(3):402-10
48. Agbla SC, De Stavola B, DiazOrdaz K. Estimating cluster-level local average treatment effects in cluster randomised trials with non-adherence. Stat Methods Med Res. 2020 Mar;29(3):911-33
49. Morden JP, Lambert PC, Latimer N, Abrams KR, Wailoo AJ. Assessing methods for dealing with treatment switching in randomised controlled trials: a simulation study. BMC Med Res Methodol. 2011 Jan 11;11:4
50. Latimer NR, Henshall C, Siebert U, Bell H. TREATMENT SWITCHING: STATISTICAL AND DECISION-MAKING CHALLENGES AND APPROACHES. International Journal of Technology Assessment in Health Care. 2016;32(3):160-6
51. Allison A, White IR, Bond S. rpsftm: An R Package for Rank Preserving Structural Failure Time Models. R j. 2017 Dec 4;9(2):342-53
52. Mansournia MA, Etminan M, Danaei G, Kaufman JS, Collins G. Handling time varying confounding in observational research. BMJ. 2017;359:j4587
53. Ellis AR, Brookhart MA. Approaches to inverse-probability-of-treatment–weighted estimation with concurrent treatments. Journal of Clinical Epidemiology. 2013 2013/08/01/;66(8, Supplement):S51-S6
54. Austin PC, Stuart EA. Moving towards best practice when using inverse probability of treatment weighting (IPTW) using the propensity score to estimate causal treatment effects in observational studies. Stat Med. 2015 Dec 10;34(28):3661-79

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APPENDIX

The Ovid MEDLINE® database was searched for terms related to treatment adherence, statistical methods for handling non-adherence, and non-inferiority trials in the titles, abstracts and key words of articles published up to 31st December 2020. The full search strategy was as follows:

1. ("intention to treat" or "intention-to-treat" or "ITT" or "intent-to-treat" or "intent to treat").ti,ab,kw.
2. ("as treated" or "as-treated").ti,ab,kw.
3. ("per protocol" or "per-protocol").ti,ab,kw.
4. ("complier average causal effect" or "CACE" or "local average treatment effect").ti,ab,kw.
5. ("non-compliance" or "noncompliance" or "compliance" or "non-adherence" or "nonadherence" or "adherence").ti,ab,kw.
6. ("instrumental variable" or "instrumental variables" or "IV analysis" or "IV analyses").ti,ab,kw.
7. ("non-inferiority" or "noninferiority" or "non-inferior" or "noninferior").ti,ab,kw.
8. 1 or 2 or 3 or 4 or 5 or 6
9. 7 and 8

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Table A1. Comparison of non-inferiority conclusions from results papers applying statistical methods of interest versus an alternate approach

Reference	Method of interest	Was method of interest applied in primary or sensitivity analysis?	Alternate analyses performed	Comparison of non-inferiority conclusions
Geldsetzer et al. 2018[1]	IV approach	Sensitivity	ITT (primary analysis, all randomised patients)	Different measures of effect
Huang et al. 2018[2]	IV approach	Sensitivity. Applied to PP analysis (patients for whom the trial intervention was achieved at all time points).	ITT (primary analysis, not defined)	Consistent – both concluded non-inferiority
Beaver et al. 2017[3]	IV approach	Primary	ITT (not defined), PP (those who had their first post-randomisation appointment in line with the randomisation), and AT (not defined)	Consistent – all concluded non-inferiority
Kitchener et al. 2006[4]	IV approach	Sensitivity	ITT (primary analysis, not defined)	Consistent – both concluded non-inferiority
Halpern et al. 2020[5]	IV approach	Sensitivity	mITT (primary analysis, those who were randomised, were not subsequently found to be ineligible, and did not withdraw consent)	Different measures of effect
Butler et al. 2019[6]	IV approach	Primary	mITT (those who had undergone randomisation and had available outcome data, regardless of protocol deviations or the intervention they received)	Consistent – both concluded non-inferiority
Bilimoria et al. 2016[7]	IV approach	Sensitivity	ITT (primary analysis, not defined), PP (limited to adherent programs), and AT (which assessed actual exposure to policy change)	Different measures of effect

Reference	Method of interest	Was method of interest applied in primary or sensitivity analysis?	Alternate analyses performed	Comparison of non-inferiority conclusions
Rimoin et al. 2011[8]	Adjustment for observed adherence	Not stated. Applied to ITT (all randomised patients meeting the inclusion and exclusion criteria) and PP analyses (those who were adherent and returned for the follow-up visit).	Unadjusted ITT and unadjusted PP	Consistent – all adjusted analyses resulted in the same conclusions as the corresponding unadjusted analyses
Hahn et al. 2019[9]	Adherence modelled as a time-varying covariate in a time-to-event analysis	Sensitivity	ITT (primary analysis, all randomised patients according to original group allocation) and PP (excluded patients who did not receive the assigned intervention)	Different measures of effect
Dignass et al. 2009[10]	Adherence modelled as a time-varying covariate in a time-to-event analysis	Sensitivity	ITT (all randomised patients who received some intervention and had at least 1 post-baseline efficacy assessment), PP1 (those who dropped out of the study censored at the time of dropout) and PP2 (excluded dropouts)	Results not provided in full
Brunori et al. 2007[11]	Adherence modelled as a time-varying covariate in a time-to-event analysis	Sensitivity	ITT (patients were considered part of the diet group even after a switch to dialysis) and PP (diet group patients were censored when switched to dialysis)	Different measures of effect
Flum et al. 2020[12]	Inverse-probability-of-treatment weighting	Sensitivity. Applied to PP analysis (those adherent to the randomisation assignment).	ITT (primary analysis, not defined)	Consistent – both concluded non-inferiority

CACE = complier average causal effect; IV = instrumental variable; ITT = intention-to-treat; PP = per-protocol; AT = as-treated; mITT = modified intention-to-treat.

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REFERENCES

1. Geldsetzer P, Francis JM, Sando D, Asmus G, Lema IA, Mboggo E, et al. Community delivery of antiretroviral drugs: A non-inferiority cluster-randomized pragmatic trial in Dar es Salaam, Tanzania. PLoS Med. 2018 Sep;15(9):e1002659
2. Huang DT, Yealy DM, Filbin MR, Brown AM, Chang CH, Doi Y, et al. Procalcitonin-Guided Use of Antibiotics for Lower Respiratory Tract Infection. N Engl J Med. 2018 Jul 19;379(3):236-49
3. Beaver K, Williamson S, Sutton C, Hollingworth W, Gardner A, Allton B, et al. Comparing hospital and telephone follow-up for patients treated for stage-I endometrial cancer (ENDCAT trial): a randomised, multicentre, non-inferiority trial. Bjog. 2017 Jan;124(1):150-60
4. Kitchener HC, Dunn G, Lawton V, Reid F, Nelson L, Smith AR. Laparoscopic versus open colposuspension--results of a prospective randomised controlled trial. Bjog. 2006 Sep;113(9):1007-13
5. Halpern SD, Small DS, Troxel AB, Cooney E, Bayes B, Chowdhury M, et al. Effect of Default Options in Advance Directives on Hospital-Free Days and Care Choices Among Seriously Ill Patients: A Randomized Clinical Trial. JAMA Netw Open. 2020 Mar 2;3(3):e201742
6. Butler CC, Gillespie D, White P, Bates J, Lowe R, Thomas-Jones E, et al. C-Reactive Protein Testing to Guide Antibiotic Prescribing for COPD Exacerbations. N Engl J Med. 2019 Jul 11;381(2):111-20
7. Bilimoria KY, Chung JW, Hedges LV, Dahlke AR, Love R, Cohen ME, et al. National Cluster-Randomized Trial of Duty-Hour Flexibility in Surgical Training. N Engl J Med. 2016 Feb 25;374(8):713-27
8. Rimoin AW, Hoff NA, Fischer Walker CL, Hamza HS, Vince A, Abdel Rahman N, et al. Treatment of streptococcal pharyngitis with once-daily amoxicillin versus intramuscular benzathine penicillin G in low-resource settings: a randomized controlled trial. Clin Pediatr (Phila). 2011 Jun;50(6):535-42
9. Hahn JY, Song YB, Oh JH, Chun WJ, Park YH, Jang WJ, et al. Effect of P2Y12 Inhibitor Monotherapy vs Dual Antiplatelet Therapy on Cardiovascular Events in Patients Undergoing Percutaneous Coronary Intervention: The SMART-CHOICE Randomized Clinical Trial. Jama. 2019 Jun 25;321(24):2428-37
10. Dignass AU, Bokemeyer B, Adamek H, Mross M, Vinter-Jensen L, Börner N, et al. Mesalamine once daily is more effective than twice daily in patients with quiescent ulcerative colitis. Clin Gastroenterol Hepatol. 2009 Jul;7(7):762-9
11. Brunori G, Viola BF, Parrinello G, De Biase V, Como G, Franco V, et al. Efficacy and safety of a very-low-protein diet when postponing dialysis in the elderly: a prospective randomized multicenter controlled study. Am J Kidney Dis. 2007 May;49(5):569-80
12. Flum DR, Davidson GH, Monsell SE, Shapiro NI, Odom SR, Sanchez SE, et al. A Randomized Trial Comparing Antibiotics with Appendectomy for Appendicitis. N Engl J Med. 2020 Nov 12;383(20):1907-19

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PRISMA 2020 Main Checklist

Topic	No.	Item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Title (page 1), abstract (page 1)
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist	
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Paragraphs 7 and 8 of introduction section (page 6)
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Paragraph 8 of introduction section (page 6)
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Paragraph 1 of methods section (pages 6 and 7)
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Paragraph 1 of methods section (pages 6 and 7)
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Full search strategy provided in the appendix
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Paragraph 1 of methods section (pages 6 and 7)
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Paragraph 1 of methods section (pages 6 and 7)

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Topic	No.	Item	Location where item is reported
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Paragraph 2 of methods section (pages 7 and 8)
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Paragraph 2 of methods section (pages 7 and 8)
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	N/A
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	N/A
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item 5)).	N/A
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	N/A
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	N/A
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Software package (paragraph 3 of methods section on page 8)
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	N/A
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	N/A
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	N/A

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Topic	No.	Item	Location where item is reported
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	N/A
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Paragraph 1 of results section (page 8) and Figure 1 (page 16)
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Paragraph 1 of results section (page 8) and Figure 1 (page 16)
Study characteristics	17	Cite each included study and present its characteristics.	Table 1 (pages 17 and 18)
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	N/A
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	N/A
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	N/A
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	N/A
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	N/A
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	N/A
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	N/A
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	N/A

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Topic	No.	Item	Location where item is reported
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Paragraph 3 of discussion section (page 11)
	23b	Discuss any limitations of the evidence included in the review.	N/A
	23c	Discuss any limitations of the review processes used.	Strengths and limitations section (page 14)
	23d	Discuss implications of the results for practice, policy, and future research.	Paragraph 7 of discussion section (page 13)
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Abstract (page 2) and paragraph 3 of methods section (page 8)
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Abstract (page 2) and paragraph 3 of methods section (page 8)
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Funding section (page 15)
Competing interests	26	Declare any competing interests of review authors.	Competing interests section (page 15)
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Data sharing statement (page 15)

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PRIMSA Abstract Checklist

Topic	No.	Item	Reported?
TITLE			
Title	1	Identify the report as a systematic review.	Yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Yes
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Yes
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	N/A
Synthesis of results	6	Specify the methods used to present and synthesize results.	N/A
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Yes
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Yes
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	No
Interpretation	10	Provide a general interpretation of the results and important implications.	Yes
OTHER			
Funding	11	Specify the primary source of funding for the review.	No
Registration	12	Provide the register name and registration number.	Yes

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From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. MetaArXiv. 2020, September 14. DOI: 10.31222/osf.io/v7gm2. For more information, visit: www.prisma-statement.org

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International prospective register of systematic reviews

NHS National Institute for Health Research

UNIVERSITY of York
Centre for Reviews and Dissemination

Systematic review

1. * Review title.

Give the working title of the review, for example the one used for obtaining funding. Ideally the title should state succinctly the interventions or exposures being reviewed and the associated health or social problems. Where appropriate, the title should use the PI(E)COS structure to contain information on the Participants, Intervention (or Exposure) and Comparison groups, the Outcomes to be measured and Study designs to be included.

Statistical methods used to handle non-adherence to treatment in non-inferiority trials: a systematic review

2. Original language title.

For reviews in languages other than English, this field should be used to enter the title in the language of the review. This will be displayed together with the English language title.

3. * Anticipated or actual start date.

Give the date when the systematic review commenced, or is expected to commence.

06/04/2020

4. * Anticipated completion date.

Give the date by which the review is expected to be completed.

03/04/2023

5. * Stage of review at time of this submission.

Indicate the stage of progress of the review by ticking the relevant Started and Completed boxes. Additional information may be added in the free text box provided.
Please note: Reviews that have progressed beyond the point of completing data extraction at the time of initial registration are not eligible for inclusion in PROSPERO. Should evidence of incorrect status and/or completion date being supplied at the time of submission come to light, the content of the PROSPERO record will be removed leaving only the title and named contact details and a statement that inaccuracies in the stage of the review date had been identified.
This field should be updated when any amendments are made to a published record and on completion and publication of the review. If this field was pre-populated from the initial screening questions then you are not able to edit it until the record is published.

The review has not yet started: No

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Review stage	Started	Completed
Preliminary searches	Yes	No
Piloting of the study selection process	No	No
Formal screening of search results against eligibility criteria	No	No
Data extraction	No	No
Risk of bias (quality) assessment	No	No
Data analysis	No	No

Provide any other relevant information about the stage of the review here (e.g. Funded proposal, protocol not yet finalised).

6. * Named contact.

The named contact acts as the guarantor for the accuracy of the information presented in the register record.

Matthew Dodd

Email salutation (e.g. "Dr Smith" or "Joanne") for correspondence:

Mr Dodd

7. * Named contact email.

Give the electronic mail address of the named contact.

matthew.dodd@lshtm.ac.uk

8. Named contact address

Give the full postal address for the named contact.

LSHTM, Keppel Street, London, WC1E 7HT

9. Named contact phone number.

Give the telephone number for the named contact, including international dialling code.

+44 20 7927 2376

10. * Organisational affiliation of the review.

Full title of the organisational affiliations for this review and website address if available. This field may be completed as 'None' if the review is not affiliated to any organisation.

London School of Hygiene & Tropical Medicine

Organisation web address:

www.lshtm.ac.uk

11. * Review team members and their organisational affiliations.

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Give the personal details and the organisational affiliations of each member of the review team. Affiliation refers to groups or organisations to which review team members belong. **NOTE: email and country are now mandatory fields for each person.**

Mr Matthew Dodd. London School of Hygiene & Tropical Medicine
Professor Diana Elbourne. London School of Hygiene & Tropical Medicine
Professor Katherine Fielding. London School of Hygiene & Tropical Medicine
Professor James Carpenter. London School of Hygiene & Tropical Medicine

12. * Funding sources/sponsors.

Give details of the individuals, organizations, groups or other legal entities who take responsibility for initiating, managing, sponsoring and/or financing the review. Include any unique identification numbers assigned to the review by the individuals or bodies listed.

No funding.

Grant number(s)

Not applicable.

13. * Conflicts of interest.

List any conditions that could lead to actual or perceived undue influence on judgements concerning the main topic investigated in the review.

None

14. Collaborators.

Give the name and affiliation of any individuals or organisations who are working on the review but who are not listed as review team members. **NOTE: email and country are now mandatory fields for each person.**

15. * Review question.

State the question(s) to be addressed by the review, clearly and precisely. Review questions may be specific or broad. It may be appropriate to break very broad questions down into a series of related more specific questions. Questions may be framed or refined using PI(E)COS where relevant.

To identify statistical methods used in non-inferiority trials that account for the impact of non-adherence to treatment.

16. * Searches.

State the sources that will be searched. Give the search dates, and any restrictions (e.g. language or publication period). Do NOT enter the full search strategy (it may be provided as a link or attachment.)

The search strategy will include terms related to patient adherence to treatment, statistical methods for handling non-adherence to treatment, non-inferiority trials, and relevant MeSH headings. The search will be performed using the Ovid MEDLINE® database. In addition, experts within the field will be consulted in order to identify key papers and the reference lists of these key papers searched. Potentially relevant manuscripts referencing the key papers will also be identified.

17. URL to search strategy.

Give a link to a published pdf/word document detailing either the search strategy or an example of a search

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strategy for a specific database if available (including the keywords that will be used in the search strategies), or upload your search strategy.Do NOT provide links to your search results.

Alternatively, upload your search strategy to CRD in pdf format. Please note that by doing so you are consenting to the file being made publicly accessible.

Do not make this file publicly available until the review is complete

18. * Condition or domain being studied.

Give a short description of the disease, condition or healthcare domain being studied. This could include health and wellbeing outcomes.

Non-adherence to treatment in non-inferiority trials.

19. * Participants/population.

Give summary criteria for the participants or populations being studied by the review. The preferred format includes details of both inclusion and exclusion criteria.

Inclusion criteria:

1. Randomised controlled trials (RCTs) with a primary analysis for non-inferiority that have applied (or plan to apply) statistical methods in order to account for the impact of non-adherence to treatment

2. Alternatively, any meta-analysis or systematic review that incorporates at least one RCT described within inclusion criteria 1, or any statistical methodology paper with a non-inferiority trial application

3. Published in English

Exclusion criteria:

1. None

20. * Intervention(s), exposure(s).

Give full and clear descriptions or definitions of the nature of the interventions or the exposures to be reviewed.

The review will include all eligible trials regardless of the intervention examined.

21. * Comparator(s)/control.

Where relevant, give details of the alternatives against which the main subject/topic of the review will be compared (e.g. another intervention or a non-exposed control group). The preferred format includes details of both inclusion and exclusion criteria.

The review will include all eligible trials regardless of the comparator to the intervention.

22. * Types of study to be included.

Give details of the types of study (study designs) eligible for inclusion in the review. If there are no restrictions on the types of study design eligible for inclusion, or certain study types are excluded, this should be stated. The preferred format includes details of both inclusion and exclusion criteria.

All RCTs with a non-inferiority design that have applied (or plan to apply) statistical methods in order to account for the impact of non-adherence to treatment. Meta-analyses and systematic reviews that incorporate such trials will also be included, as well as statistical methodology papers with a non-inferiority trial application.

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23. Context.

Give summary details of the setting and other relevant characteristics which help define the inclusion or exclusion criteria.

The review will not be restricted to any particular geographical setting, disease area, journal(s), or publication period. The review will focus on identifying relevant statistical methods, how the methods impact on the trial results/conclusions, and their advantages and disadvantages.

24. * Main outcome(s).

Give the pre-specified main (most important) outcomes of the review, including details of how the outcome is defined and measured and when these measurement are made, if these are part of the review inclusion criteria.

1. What was the statistical method applied in order to account for non-adherence to treatment and what was the estimand (the parameter estimated)? Alternatively, for protocol papers, what statistical method is planned to be applied in order to account for non-adherence to treatment?

2. What was the impact on the trial results/conclusions compared with those from the primary analysis?

3. What are the advantages and disadvantages of the methods?

* Measures of effect

Please specify the effect measure(s) for you main outcome(s) e.g. relative risks, odds ratios, risk difference, and/or 'number needed to treat.

Not applicable.

25. * Additional outcome(s).

List the pre-specified additional outcomes of the review, with a similar level of detail to that required for main outcomes. Where there are no additional outcomes please state 'None' or 'Not applicable' as appropriate to the review

1. The average level of non-adherence to treatment in the non-inferiority trials included.

* Measures of effect

Please specify the effect measure(s) for you additional outcome(s) e.g. relative risks, odds ratios, risk difference, and/or 'number needed to treat.

Average proportion of non-adherence to treatment.

26. * Data extraction (selection and coding).

Describe how studies will be selected for inclusion. State what data will be extracted or obtained. State how this will be done and recorded.

The titles and abstracts retrieved using the search strategy will be compared against the eligibility criteria and then the full texts of potentially relevant manuscripts reviewed. This process will be performed independently by two reviewers. Any discrepancies will be discussed with a third reviewer in order to reach a consensus regarding the studies to be included. A standardised electronic form will be used to extract the relevant information from each manuscript included in the review. This will include:

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Health Research

Trial characteristics

• Author

• Publication year

• Title

• Journal

• Disease area

• Type of intervention (surgery, drug, device, other)

• Sample size

Primary analysis

• Description of primary endpoint

• Type of primary endpoint (binary, continuous, count, time-to-event)

• Analysis population for primary analysis (intention-to-treat, PP, AT, other)

• Estimand in the primary analysis

• Non-inferiority margin

• Estimated effect of intervention in primary analysis (with confidence interval (CI))

• Significance level of the CI

• One sided or two-sided CI

• Conclusion regarding effect of intervention from primary analysis (non-inferior, inferior, superior, undetermined)

Non-adherence to treatment

• Definition of non-adherence used

• All-or-nothing or continuous measure of adherence

• How adherence was measured

• Estimated level of non-adherence to treatment

Sensitivity analysis

• Statistical method used to account for the impact of non-adherence to treatment

• Estimand in the sensitivity analysis

• Estimated effect of intervention in sensitivity analysis (with CI)

• Conclusion regarding effect of intervention from sensitivity analysis (non-inferior, inferior, superior, undetermined)

• Advantages of the statistical method applied as stated by the authors

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• Disadvantages of the statistical method applied as stated by the authors

27. * Risk of bias (quality) assessment.

Describe the method of assessing risk of bias or quality assessment. State which characteristics of the studies will be assessed and any formal risk of bias tools that will be used.

The focus of the review is on identifying statistical methods that have been (or will be) applied and not on estimating the effect of an intervention. The exclusion of manuscripts not published in English is a potential source of bias.

28. * Strategy for data synthesis.

Provide details of the planned synthesis including a rationale for the methods selected. This **must not be generic text** but should be **specific to your review** and describe how the proposed analysis will be applied to your data.

A summary table of the relevant non-inferiority trials will be included along with a description of each statistical method identified and a discussion of their advantages and disadvantages.

29. * Analysis of subgroups or subsets.

State any planned investigation of 'subgroups'. Be clear and specific about which type of study or participant will be included in each group or covariate investigated. State the planned analytic approach.

None.

30. * Type and method of review.

Select the type of review and the review method from the lists below. Select the health area(s) of interest for your review.

Type of review

Cost effectiveness
No

Diagnostic
No

Epidemiologic
No

Individual patient data (IPD) meta-analysis
No

Intervention
No

Meta-analysis
No

Methodology
No

Narrative synthesis
No

Network meta-analysis
No

Pre-clinical
No

Prevention

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No

Prognostic
No

Prospective meta-analysis (PMA)
No

Review of reviews
No

Service delivery
No

Synthesis of qualitative studies
No

Systematic review
No

Other
No

Health area of the review

Alcohol/substance misuse/abuse
No

Blood and immune system
No

Cancer
No

Cardiovascular
No

Care of the elderly
No

Child health
No

Complementary therapies
No

Crime and justice
No

Dental
No

Digestive system
No

Ear, nose and throat
No

Education
No

Endocrine and metabolic disorders
No

Eye disorders
No

General interest
No

Genetics
No

Health inequalities/health equity

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NHS National Institute for Health Research

No

Infections and infestations
No

International development
No

Mental health and behavioural conditions
No

Musculoskeletal
No

Neurological
No

Nursing
No

Obstetrics and gynaecology
No

Oral health
No

Palliative care
No

Perioperative care
No

Physiotherapy
No

Pregnancy and childbirth
No

Public health (including social determinants of health)
No

Rehabilitation
No

Respiratory disorders
No

Service delivery
No

Skin disorders
No

Social care
No

Surgery
No

Tropical Medicine
No

Urological
No

Wounds, injuries and accidents
No

Violence and abuse
No

31. Language.

Select each language individually to add it to the list below, use the bin icon to remove any added in error.

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English

There is not an English language summary

32. * Country.

Select the country in which the review is being carried out from the drop down list. For multi-national collaborations select all the countries involved.

England

33. Other registration details.

Give the name of any organisation where the systematic review title or protocol is registered (such as with The Campbell Collaboration, or The Joanna Briggs Institute) together with any unique identification number assigned. (N.B. Registration details for Cochrane protocols will be automatically entered). If extracted data will be stored and made available through a repository such as the Systematic Review Data Repository (SRDR), details and a link should be included here. If none, leave blank.

34. Reference and/or URL for published protocol.

Give the citation and link for the published protocol, if there is one

Give the link to the published protocol.

Alternatively, upload your published protocol to CRD in pdf format. Please note that by doing so you are consenting to the file being made publicly accessible.

No I do not make this file publicly available until the review is complete

Please note that the information required in the PROSPERO registration form must be completed in full even if access to a protocol is given.

35. Dissemination plans.

Give brief details of plans for communicating essential messages from the review to the appropriate audiences.

The review team intend to publish this systematic review in a peer-reviewed journal.

Do you intend to publish the review on completion?

Yes

36. Keywords.

Give words or phrases that best describe the review. Separate keywords with a semicolon or new line. Keywords will help users find the review in the Register (the words do not appear in the public record but are included in searches). Be as specific and precise as possible. Avoid acronyms and abbreviations unless these are in wide use.

systematic review; statistical methods; methodology; non-adherence; adherence; non-inferiority; trial

37. Details of any existing review of the same topic by the same authors.

Give details of earlier versions of the systematic review if an update of an existing review is being registered, including full bibliographic reference if possible.

38. * Current review status.

Review status should be updated when the review is completed and when it is published. For

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newregistrations the review must be Ongoing.
Please provide anticipated publication date

Review_Ongoing

39. Any additional information.

Provide any other information the review team feel is relevant to the registration of the review.

40. Details of final report/publication(s).

This field should be left empty until details of the completed review are available.

Give the link to the published review.

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BMJ Open

Statistical Methods for Non-adherence in Non-inferiority Trials: Useful and Used? A Systematic Review

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-052656.R1
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Primary Subject Heading:	Research methods
Secondary Subject Heading:	Research methods
Keywords:	STATISTICS & RESEARCH METHODS, GENERAL MEDICINE (see Internal Medicine), PUBLIC HEALTH

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STATISTICAL METHODS FOR NON-ADHERENCE IN NON-INFERIORITY TRIALS: USEFUL AND USED? A SYSTEMATIC REVIEW

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ABSTRACT

Background: In non-inferiority trials with non-adherence to interventions (or non-compliance), intention-to-treat and per-protocol analyses are often performed; however non-random non-adherence biases estimates of efficacy.

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Objective: To identify statistical methods that adjust for the impact of non-adherence and thus estimate the causal effects of experimental interventions in non-inferiority trials.

Design: A systematic review was conducted by searching the Ovid MEDLINE® database (31st December 2020) to identify 1) randomised trials with a primary analysis for non-inferiority that applied (or planned to apply) statistical methods to account for the impact of non-adherence to interventions, and 2) methodology papers that described such statistical methods and included a non-inferiority trial application.

Outcomes: The statistical methods identified, their impacts on non-inferiority conclusions, and their advantages/disadvantages.

Results: A total of 24 papers were included (4 protocols, 13 results papers, 7 methodology papers) reporting relevant methods on 26 occasions. The most common were instrumental variable approaches (n=9), including observed adherence as a covariate within a regression model (n=3), and modelling adherence as a time-varying covariate in a time-to-event analysis (n=3). Other methods included rank preserving structural failure time models and inverse-probability-of-treatment weighting. The methods identified in protocols and results papers were more commonly specified as sensitivity analyses (n=13) than primary analyses (n=3). Twelve results papers included an alternative analysis of the same outcome; conclusions regarding non-inferiority were in agreement on six occasions and could not be compared on six occasions (different measures of effect or results not provided in full).

Conclusions: Available statistical methods which attempt to account for the impact of non-adherence to interventions were used infrequently. Therefore, firm inferences about their influence on non-inferiority conclusions could not be drawn. Since intention-to-treat and per-protocol analyses do not guarantee unbiased conclusions regarding non-inferiority, the methods identified should be considered for use in sensitivity analyses.

Study registration: Prospero CRD42020177458.

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ARTICLE SUMMARY

Strengths and limitations of this study

- This is the first systematic review to identify statistical methods that attempt to account for the impact of non-adherence to interventions in randomised non-inferiority trials.
- A description and critique of the statistical methods identified is provided, along with their target estimands.
- Publications from any year, journal, or disease area/patient population were reviewed independently by two authors.
- One author extracted the data from the eligible papers.
- Whilst statistical analysis plans were requested for eligible trials, these could not be obtained for all included trials.

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INTRODUCTION

Non-inferiority trials, which assess whether a new intervention is not worse than a proven comparator by more than a clinically acceptable amount, are becoming increasingly common.[1-3] They are principally used when it is hoped that the new intervention may convey some advantage other than better efficacy (its effect under ideal conditions), such as improved safety, tolerability, convenience, or reduced cost.[4, 5]

One of the challenges in these studies, and the focus of this review, is how participants not receiving their randomly assigned intervention according to the trial protocol (termed non-adherence or non-compliance) should be handled in the statistical analysis.[6] Examples of non-adherence include not receiving a surgical intervention as planned, not taking all of the prescribed doses of a medication, or not attending all of the sessions of an exercise rehabilitation program. Such non-adherence is common in trials and has been associated with poorer health outcomes.[7-9] It can bias estimates of efficacy in either direction and so obtaining an accurate and reliable measure of adherence and accounting for any non-adherence in the statistical analysis of these studies is essential.[10, 11] Non-adherence may also be linked with missing outcome data if, for example, the trial protocol stipulates that further follow-up is no longer required once adherence drops below a specific threshold or if non-adherent participants become lost to follow-up. The terms adherence and compliance are often used interchangeably, though adherence is preferred here since it is felt to better reflect the partnership between the healthcare provider and participant.

A simple approach to handling non-adherence is to define and analyse different analysis sets based on participants' observed levels of adherence, with consistent results providing greater confidence in the trial conclusions.[1] In the setting of non-inferiority trials, the intention-to-treat (ITT) and per-protocol (PP) populations have been advocated and are commonly used.[4, 12, 13] However, agreement between the ITT and PP results of these trials does not guarantee that conclusions regarding non-

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inferiority are free from bias caused by differential, or non-random, non-adherence (where the factors leading to non-adherence are associated with outcomes).[14-16]

Standard ITT analyses typically include all participants in their randomised groups irrespective of the intervention actually received.[17] Thus, they reflect the effect of assigning individuals to interventions in clinical practice where not everyone is fully adherent (also known as the 'effectiveness' of an intervention). This approach preserves the balance in known and unknown prognostic factors afforded by randomisation and so any difference in outcomes between study arms can be attributed solely to the experimental intervention.[18] However, in the presence of non-adherence, ITT analyses may yield biased estimates of efficacy (also known as the 'causal effect' of an intervention).[19] In non-inferiority trials, where efficacy and effectiveness may be considered equally important, this can increase the probability of falsely claiming non-inferiority and therefore accepting a worse intervention.[11]

Modified intention-to-treat (mITT) analyses are commonly used to address some of the limitations with standard ITT methods.[20] This approach allows some randomised participants, such as those who never receive any of the allocated intervention or who are identified as ineligible after randomisation, to be excluded according to pre-specified rules.[18] However, across trials, there is substantial variability in how this population is defined and bias may be introduced by subjectively excluding individuals from analysis.[18, 20] In addition, mITT analyses are not typically used to account for the impact of non-adherence.

PP analyses estimate the efficacy of interventions typically by excluding or censoring individuals with major protocol violations, including those who are non-adherent to their allocated intervention.[1, 6, 17] Excluding participants in this way can lead to selection bias because non-adherent individuals generally differ from those who are fully adherent with respect to prognostic factors.[21, 22] Furthermore, using a PP analysis to address differential non-adherence is likely to reduce the

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protection provided by randomisation, so that trial arms are not fully comparable; this potentially biases the study results in either direction.[11] In other words, any difference in outcomes between trial arms may no longer be due to the experimental intervention only. To obtain valid results from a PP analysis, we need to recover the protection due to randomisation, typically through a statistical method that (given certain assumptions) correctly adjusts for factors associated with both adherence and outcome (confounders).[21]

Statistical techniques that attempt to account for the impact of non-adherence and thus estimate the causal effects of experimental interventions exist. These range from simple approaches, such as including observed adherence as a covariate within a regression model, which like PP analyses is susceptible to selection bias, to more sophisticated techniques, such as instrumental variable (IV) methods and inverse-probability weighting, which allow for non-adherence while attempting to maintain the balance produced by randomisation.[23, 24] Several of these methods attempt to estimate the complier average causal effect (CACE), which is the causal effect of an intervention for individuals who would always be fully adherent regardless of assignment (known as compliers).[25] In other words, it is a comparison of the average outcome among those who are fully adherent in the experimental arm with the average outcome among the comparable group in the control arm who would fully adhere to the experimental intervention, if offered.

It is unclear which of the alternative methods have been applied in the setting of non-inferiority trials, to what extent, and with what results. Therefore, this systematic review aimed to identify statistical methods that can be used to account for the impact of non-adherence to interventions (thereby estimating the causal effects of experimental interventions) in randomised non-inferiority trials. Secondary aims were to quantify the use of such methods in these studies and examine their impact on non-inferiority conclusions.

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METHODS

The Ovid MEDLINE® database was searched for terms related to adherence, non-inferiority trials, and statistical methods for handling non-adherence in the titles, abstracts and key words of papers published up to 31st December 2020 (full search strategy provided in the appendix). Eligibility based on identifying appropriate statistical methods was assessed using a three-stage process. First, two authors independently reviewed the title and abstract of each paper. Those where the comparison was not randomised, the primary analysis was not for non-inferiority, or the analysis assessed cost-effectiveness were excluded (cost-effectiveness analyses were not of interest because the focus of this review was on estimating the efficacy of interventions). Papers not published in English were also excluded. Even if the full text was unavailable, the abstract was reviewed against the eligibility criteria to ensure that key papers were not excluded. Next, an automated search of the full texts was performed in order to identify those containing the terms 'compliance', 'adherence' or 'complier'. Finally, full-text reviews of the remaining papers were performed independently by two authors to identify, 1) randomised trials with a primary analysis for non-inferiority that applied (or planned to apply, for protocol papers) statistical methods to account for the impact of non-adherence to interventions, and 2) methodology papers that described such statistical methods and included a non-inferiority trial application. Any discrepancies between reviewer pairs were discussed with a third author in order to reach a consensus. In addition, statisticians within the field were consulted in order to identify key publications, and the reference lists and citations of eligible papers searched for relevant analyses (performed by one author (MD)). Meta-analyses and systematic reviews identified were also searched for eligible non-inferiority trials. Where a trial's published protocol and results paper were both eligible and reported the same statistical method of interest, the protocol paper was excluded to avoid double counting. Statistical analysis plans were requested for all eligible trials.

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A standardised electronic form was used to extract the relevant information from each paper considered eligible. This included details of the trial characteristics (journal, year of publication, disease area or patient population, unit of randomisation, type of experimental intervention, type of primary outcome, and non-inferiority margin), non-adherence to the interventions (definitions and estimated levels of non-adherence), the statistical method attempting to account for non-adherence (name of the method, estimand, estimate of effect and confidence interval, conclusion regarding non-inferiority, and any advantages/disadvantages of the method stated), and any other analyses applied to the same outcome (analysis population, estimand, estimate of effect and confidence interval, and conclusion regarding non-inferiority). Data extraction was performed by one author (MD). The primary outcome was the statistical method applied (or planned to be applied) in order to account for non-adherence to the interventions. Other outcomes were the impact of applying these methods on the trial conclusions (compared with other analyses applied to the same outcome, where available) and the advantages and disadvantages of the methods where stated by the authors. The impact of applying the methods of interest was assessed using trial results papers only.

This systematic review was registered with Prospero (registration CRD42020177458) and conducted in accordance with the PRISMA statement.[26] Information was largely combined using a narrative synthesis approach i.e. "synthesis of findings from multiple studies that relies primarily on the use of words and text to summarise and explain the findings of the synthesis".[27] All analyses were conducted using Stata version 15.1.

Ethics approval

Ethical approval was not required because this review used publicly accessible documents.

Patient and public involvement

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Patients or the public were not involved in the design, conduct, reporting, or dissemination plans of our research.

RESULTS

After removing duplicate publications, our search identified 3,235 papers. Of these, 934 were excluded following review of the titles and abstracts, 790 did not contain any key words in the full texts, and 1,489 were excluded after full-text review, leaving 22 papers whose citations and reference lists contained a further five papers meeting eligibility criteria. After removing publications of the same trial reporting identical statistical methods of interest, 24 papers remained (Figure 1).

The 24 publications, which consisted of four protocols, 13 results papers, and seven methodology papers, reported relevant methods on 26 occasions (two methodology papers both contained two relevant analyses). Four of the analyses included in methodology papers were re-analyses of non-inferiority trials, one included a simulation study based on a non-inferiority trial, and four included simulation studies not based on real trials. Fifteen of the 24 papers included (63%) were published within the last five years and the most common type of experimental intervention studied was drug interventions (35%) (Table 1 and Table A1 in the appendix).

Non-adherence to interventions

Non-adherence to the randomly assigned interventions was defined in the methods, statistical analysis plan, or results section of most analyses (n=19, 73%). Fifteen (79%) used a binary definition of adherence, whereas three (16%) used a continuous measure (one was unclear). Of the 19 analyses that defined non-adherence to the interventions, 13 reported estimates of non-adherence (the remaining six were protocols or simulation studies). More than half reported estimates of non-adherence that were no more than 10%, though the range was wide (1.7% to 51.3%) (Table 2). For

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reasons that were not reported, two papers provided data on non-adherence to the interventions in only one arm of the trial.

Statistical methods for handling non-adherence to interventions

In total, 11 different statistical methods that attempt to account for non-adherence to interventions were identified (Table 3). The most common were IV approaches (n=9, 35%), including observed adherence as a covariate within a regression model (n=3, 12%), and modelling adherence as a time-varying covariate in a time-to-event analysis (n=3, 12%). Other methods included rank preserving structural failure time models and G-estimation (n=2, 8%), inverse-probability-of-treatment weighting (n=2, 8%) and the tipping point approach (n=2, 8%, both in the same methodology paper). The other five techniques identified were all reported once. Further details of the methods reported more than once are provided in Table 4 and Table A2 in the appendix. The techniques identified in the 17 protocols and results papers were more commonly specified as sensitivity analyses (n=13, 76%) than primary analyses (n=3, 18%) (one was unclear).

Advantages and disadvantages of the statistical methods

The advantages and disadvantages of the methods identified (as stated by the authors) are shown in Table 3. Advantages or disadvantages of the techniques used were stated in eight (33%) of the 24 papers included; six were methodological papers and two were results papers. No advantages or disadvantages were stated for five of the 11 methods identified.

Impact of the statistical methods on non-inferiority conclusions

Twelve of the 13 results papers (92%) also included an alternative analysis of the same outcome (Table A3 in the appendix). All 12 performed an ITT or mITT analysis. In addition, some reported results from PP (n=6, 50%) or as-treated (AT; n=2, 17%) analyses. Non-inferiority conclusions from the alternate analyses were in agreement with those from the methods of interest on six occasions and could not

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be compared on six occasions (due to different measures of effect or the results not being provided in full). Five of the six analyses where the different methods were in agreement concluded non-inferiority of the experimental intervention versus the comparator. The remaining trial provided mixed findings regarding non-inferiority across the two different countries included, though the interpretation of this study appeared inconsistent with its design (a confidence interval approach to determining non-inferiority was stated in the methods but not utilised).

Statistical analysis plans

Statistical analysis plans were requested for all 17 non-inferiority trials where the protocol or results paper was included in the review, and obtained for nine of these trials.

DISCUSSION

To our knowledge, this is the first systematic review undertaken to both identify statistical methods that adjust for the impact of non-adherence to interventions in randomised non-inferiority trials and also identify the frequency and consequences of their use. We found that few papers reported such methods (less than 2% of those reaching full-text review). This may be partly due to unfamiliarity with such techniques amongst trialists and statisticians as a result of the long lead time for statistical methodology to make its way into routine practice. The most common techniques identified were IV approaches, including observed adherence as a covariate within a regression model, and modelling adherence as a time-varying covariate in a time-to-event analysis. Overall, the number of trials implementing relevant statistical methods was too small to draw firm inferences about their impacts on non-inferiority conclusions. In six analyses where the results from methods of interest could be compared directly with those from an alternative analysis, conclusions regarding non-inferiority were consistent across the different approaches.

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Almost half of the methods identified focus on estimating CACE (also known as the local average treatment effect (LATE)). This is the average effect of the experimental intervention within the subpopulation of compliers.[25] We argue that this is the natural estimation focus when attempting to account for non-adherence to interventions in the context of non-inferiority trials. This is because we want to be confident that there is non-inferiority among those who would comply with either intervention. By contrast, including participants who would not fully adhere to both interventions may bias estimation towards non-inferiority (in a similar way that, in the context of non-inferiority, ITT analyses may be biased towards non-inferiority under non-adherence). For similar reasons, we believe that the CACE is preferable to the population average treatment effect (ATE). Lastly, we note that when adjusting for observed adherence within a regression model or modelling adherence as a time-varying covariate in a time-to-event analysis, the target estimand is unclear.

The infrequent use of statistical methods for handling non-adherence seen in the current review has also been observed more generally in randomised controlled trials (RCTs). A review of one hundred RCTs randomly selected from those published in four high-impact journals during 2008 found only one that attempted to account for non-adherence to interventions using a causal inference framework (in which inverse-probability-of-censoring weighting was applied).[6] More recently, Mostazir et al. conducted a review of statistical approaches for handling non-adherence to interventions in RCTs published between 1991 and 2015 which identified 88 analyses incorporating nine different methods.[28] IV methods were among the most common and accounted for almost one in four applications of suitable techniques. However, some of the other methods identified (including CACE analyses using maximum-likelihood estimation and adjusted treatment received models) were not captured in the current review focussing on non-inferiority trials. Similarly, we did not identify all twelve approaches included in a recent review of methodological papers containing statistical techniques for handling non-adherence to interventions in the context of time-to-event outcomes.[29] This suggests that other relevant methods are available but either they are not suitable for comparing

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active interventions, as is often required in non-inferiority trials, or they may not have been applied within these studies. The three aforementioned reviews did not focus specifically on non-inferiority trials.

It is perhaps not surprising that IV approaches were the most common method identified in the current review, given that their assumptions are well suited to many double-blind trials, they can be applied across a range of trial designs, and they are relatively simple to implement in standard statistical software.[30] IV methods use randomisation as the instrument in order to account for unmeasured confounders of the outcome and intervention received (i.e. adherence). Their main assumptions are: 1) randomisation affects the outcome only through its influence on the intervention received (the *exclusion restriction*), 2) randomisation does not share common causes with the outcome (the *exchangeability* assumption), 3) randomisation causes some participants to receive their assigned intervention (the *relevance* assumption), and, in order to estimate CACE, 4) there are no participants who would always receive the opposite of their random allocation (the *monotonicity* assumption).[23, 31] In individually randomised trials, the exclusion restriction and monotonicity assumptions are typically satisfied by effective double-blinding and/or use of objective outcomes, and the exchangeability assumption is usually valid since randomisation is expected to produce trial arms that are balanced with respect to prognostic factors.

When these assumptions hold, it is relatively straightforward to show that if we regress the intervention received (i.e. adherence) on randomisation, and then use this model to predict each participant's adherence, these predictions are orthogonal (independent) of all adherence-outcome confounders. Therefore, if in a second step we regress the outcome on these predictions, we get an unconfounded estimate of the effect of adherence on outcome. It follows that, in contrast to techniques that involve inverse-probability weighting, when the above four IV assumptions hold, IV methods enable us to estimate CACE even in the presence of unmeasured confounding (although

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inclusion of measured confounders can improve precision).[32, 33] While IV methods may thus appear a panacea, as usual in statistics, there are no free lunches: a lack of precision and statistical power is often a challenge with IV techniques and methods used to adjust for non-adherence more generally.[5, 30, 34, 35]

The two-stage least-squares regression (2SLS) approach sketched in the previous paragraph can be applied when the intervention is not all or nothing. Suppose that a non-inferiority trial is conducted to assess whether prescribing one dose of a medication per week is non-inferior to prescribing two doses per week over the course of four weeks. For each participant, the monotonicity assumption requires that the potential number of doses taken would be lower if the participant was randomly assigned to receive one dose per week than if they were randomised to receive two doses per week. Assuming there are no covariates and the monotonicity assumption holds, it can be shown that the 2SLS estimator converges towards a weighted average of the causal effects of one unit increases in the intervention amongst compliers (individuals whose intervention intensity is affected by randomisation (the instrument)).[36, 37] This is because the implicit effect of the 2SLS analysis is that values of the outcome at which there are more compliers get given greater weight.

A limitation of IV methods is that when interventions are administered at multiple timepoints, standard approaches are susceptible to time-varying confounding and selection bias.[21] These biases occur when previous values of a covariate predict the current intervention received and the current value of the covariate predicts outcome.[38] If the time-varying confounders are themselves affected by previous intervention received, so-called G-methods, such as inverse-probability weighting or G-estimation, are required to allow for the feedback loop occurring between the intervention received and confounders over time.[21, 24, 39] G-methods were seldom reported in the current review, perhaps because they can be more complex to implement than alternative approaches and also rely on assumptions which may be vulnerable to violations. When considering whether to apply an IV

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approach or a G-method, statisticians might consider whether the exclusion restriction and monotonicity assumptions are realistic given the context of the trial, and whether randomisation is a sufficiently strong instrument. Where outcomes are collected at multiple timepoints, inverse-probability weighting may be a more attractive approach if data on potential confounders are also collected throughout follow-up.

In order to estimate the effect of the experimental intervention in the absence of (full) protection by randomisation, additional assumptions must be made. Most of these are, by their nature, inherently untestable. Each of the statistical methods identified in the current review make slightly different assumptions in order to estimate the effects of interventions under full adherence, and hence each has a different method of estimation; both assumptions and estimation methods have associated advantages and disadvantages. Crucially, all of the methods require reliable information regarding adherence to the randomly assigned interventions, which is often challenging to measure, particularly for long-term therapies.[40]

Despite these limitations, it is our view that the methods identified have an important role in non-inferiority trials with non-adherence to interventions and should be applied as sensitivity analyses alongside other techniques (such as ITT, PP and AT analyses). Given that agreement between ITT and PP analyses cannot guarantee unbiased conclusions in these studies, those with non-trivial non-adherence should assess the sensitivity of trial results to different assumptions in order to guard against falsely claiming non-inferiority and accepting a worse intervention. Careful consideration needs to be given to the assumptions that are most plausible given the trial context and planned design, before selecting the appropriate statistical method to adjust for potential non-adherence. Relevant data needed to implement the chosen technique should then be collected as fully as possible. Clearly, the best approach to reducing the potential biases introduced by non-adherence to the interventions is to design trials that minimise such non-adherence. Future work should compare the

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performance of the methods identified under different non-adherence scenarios in non-inferiority trials to facilitate understanding of when they might be applied appropriately.

Strengths and limitations

This is the first systematic review (protocol published on PROSPERO) to identify statistical methods that attempt to account for the impact of non-adherence to interventions in randomised non-inferiority trials, quantify the use of such methods in these studies, and examine their impact on non-inferiority conclusions. The review included publications from any year, journal or disease area and involved two authors agreeing the eligibility of each paper identified in the search. However, it has some limitations. First, the search for eligible papers had to be restricted to those containing terms related to adherence or statistical methods for handling non-adherence in the titles, abstracts and key words. Publications applying suitable methods in sensitivity analyses may not have referred to the techniques within these fields and, if so, would not have been captured by the search. A wide range of search terms were used to try and mitigate this problem and therefore a large number of papers reviewed. Second, only one database was searched and papers not published in English were excluded meaning it is possible that some eligible papers may not have been captured by the search. However, most major non-inferiority trials are likely to be published in English within one of the MEDLINE® journals and therefore should have been captured. Third, one author performed the data extraction from eligible publications, though other authors were consulted where necessary. Finally, whilst statistical analysis plans were requested, these could not be obtained for eight of the trials included. For these studies, we cannot be sure that the details provided in the publications reviewed are accurate accounts of the planned analyses.

CONCLUSION

In non-inferiority trials with non-adherence to interventions, ITT and PP analyses are often performed but may result in biased estimates of efficacy and therefore agreement between these approaches

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does not guarantee that conclusions regarding non-inferiority are unbiased. Statistical methods that attempt to account for the impact of non-adherence and thereby estimate the causal effects of interventions are available, but their use in non-inferiority trials remains extremely infrequent. It is our view that the methods identified should be applied more widely within sensitivity analyses of non-inferiority trials. In particular, those with non-trivial non-adherence should assess the sensitivity of trial results to different assumptions in order to guard against falsely claiming non-inferiority and accepting a worse intervention.

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Author contributions

MD devised the review, reviewed the eligibility of papers identified, carried out data extraction and analysis, and wrote the initial manuscript. KF, JC, JAT and DE commented on the review protocol, reviewed the eligibility of papers identified, and assisted with writing the final manuscript.

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Competing interests

None declared.

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Data sharing statement

No additional data are available.

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Figure 1. Flow chart showing the eligibility of papers reviewed *(uploaded separately)*

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Table 1. Characteristics of eligible analyses (n=26)

Characteristics	n (%)
Type of publication (n=24)	
Results	13 (54)
Methodology	7 (29)
Protocol	4 (17)
Year of publication (n=24)	
2006 – 2010	5 (21)
2011 – 2015	4 (17)
2016 – 2020	15 (63)
Disease area or patient population	
Mental health	4 (15)
Appendicitis	2 (8)
Cancer	2 (8)
Respiratory infection/disease	2 (8)
Ulcerative colitis	2 (8)
Anaemia	1 (4)
Ear infection	1 (4)
General surgery patients	1 (4)
Heart disease	1 (4)
Human immunodeficiency virus (HIV)	1 (4)
Individuals receiving life-sustaining therapies	1 (4)
Renal disease	1 (4)
Smoking cessation	1 (4)
Throat infection	1 (4)
Urinary incontinence	1 (4)
Simulation study	4 (15)
Unit of randomisation	
Individual	19 (73)
Cluster	3 (12)
Simulation study	4 (15)
Type of experimental intervention	
Drug	9 (35)
Method of treatment delivery	3 (12)
Additional patient examination	2 (8)
Nutritional	2 (8)
Surgical	1 (4)
Simulation study	4 (15)
Other	5 (19)
Type of outcome	
Binary	13 (50)
Continuous	8 (31)
Time-to-event	4 (15)
Count	1 (4)
Composite outcome	3 (12)

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Table 2. Estimates of non-adherence to interventions reported in methodology and results papers, combined across trial arms unless reported (n=13)

Estimate of non-adherence	Binary measure of adherence (n=11)	Continuous* measure of adherence (n=2)	Binary or continuous* measure of adherence (n=13)
≤5%	4 (36)	0 (0)	4 (31)
6 – 10%	4 (36)	0 (0)	4 (31)
11 – 25%	0 (0)	1 (50)	1 (8)
26 – 50%	2 (18)†	1 (50)	3 (23)
>50%	1 (9)†	0 (0)	1 (8)

Data presented as n (%). * Mean level of non-adherence; † Two papers provided an estimate of non-adherence in only one arm of the trial.

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Table 3. Statistical methods that were identified as attempting to account for non-adherence to interventions

Method (estimand†)	Brief description	n (%)	Advantages*	Disadvantages*
IV approaches (CACE)	CACE estimated using the conventional IV estimator, the Generalized Method of Moments IV estimator, or two-stage least squares (2SLS) regression	9 (35)	1) Straightforward to compute[30] 2) Preserves the balance in patient characteristics from randomization[30] 3) Although the validity of the IV estimator depends on several key assumptions, these assumptions are likely to hold in most double-blinded studies[30] 4) Correctly adjusts for missing outcome data, assuming that these are missing at random (MAR)[41] 5) Can account for unknown confounders[11] 6) Recent methods using doubly robust procedures have been developed to boost power when using IV estimation[11]	1) Accurate data on compliance behaviour must also be available[30] 2) The IV method… does increase the sample size requirements of the study as the expected proportion of noncompliers increases[30] 3) Requires the "exclusion restriction" to be fulfilled (i.e. treatment allocation only influences the outcome through the treatment and not through any other pathways). This assumption is unverifiable and we are only likely to be confident that it holds in a double blinded study.[11]
Adjustment for observed adherence (estimand unclear)	Observed adherence included as a covariate within a regression model	3 (12)	None stated	None stated
Adherence modelled as a time-varying covariate in a time-to-event analysis (estimand unclear)	Attributes the time at risk between observations and the outcomes occurring during the same period to the concurrent value of adherence	3 (12)	None stated	1) Results… (from the) time-dependent analysis could be the consequence of a selection bias[42]
Tipping point approach (estimates the probability of reversing the trial conclusions under a range of assumptions about the outcome data following non-adherence)	Outcome data following non-adherence is treated as missing. Assesses how sensitive the trial results are to the values of these missing outcomes.	2 (8)†	1) A model or mechanism for the missing outcomes does not have to be assumed[43] 2) All randomised individuals are included in the analysis[43]	1) Require(s) assumptions that are often difficult or not possible to verify[43]

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Method (estimand†)	Brief description	n (%)	Advantages*	Disadvantages*
Rank preserving structural failure time model and G-estimation (ATE)	Untreated survival times (those that would occur if no EXP were received) are assumed to be a function of both time on/off EXP and the effect of EXP compared with CON. The value for the effect of EXP that results in equal untreated survival times in the randomised groups is identified (via G-estimation) and used to calculate adjusted survival times that would have been observed had no switching occurred.	2 (8)	1) Takes compliance history into account[44] 2) Maintains the original randomized group[45] 3) (G-estimation) can be performed for other types of outcomes, such as continuous, binary, or count responses, using structural nested mean models[45]	1) Can only be used under a specific set of assumptions[44] 2) The rank preserving structural failure time model… incorporates a strong non-interaction assumption with respect to the treatment effect[45]
Inverse-probability-of-treatment weighting (ATE)	A pseudo-population is created by re-weighting participants' outcomes according to the probability of adherence at each visit, given previous values of the intervention received and confounders. The causal effect of EXP is estimated by performing an unadjusted analysis in the pseudo-population, which is equivalent to a weighted analysis in the original cohort.	2 (8)	1) Ensure(s) that the reweighted arms are similar and comparable[11] 2) Sensitivity analysis methods are available to address unobserved confounding and covariate measurement errors[11]	1) It eliminates bias if all confounders can be appropriately adjusted for, but in general this will not be possible[11]
Structural mean models (CACE)	Baseline variables that predict adherence differentially in each arm of the trial and are also conditionally independent of outcome are identified. Enables the estimation of two distinct causal parameters, from which a contrast can be made.	1 (4)	1) Straightforward to implement using standard statistical software[35]	1) This paper highlights the increase in variance experienced when fitting these models, something that can only be reduced when the models include strong predictors of adherence and outcome[35] 2) Extension of the approach to handle nonlinear outcome variables is also required[35]

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Method (estimand†)	Brief description	n (%)	Advantages*	Disadvantages*
CACE analysis using propensity score approach (CACE)	A propensity score is developed in the EXP group in order to predict the probability that those in the CON group would have been fully adherent if assigned to EXP. CON group outcomes are re-weighted using these probabilities and compared with the outcomes of those who were fully adherent in the EXP group.	1 (4)	None stated	None stated
CACE analysis using a mixture modelling approach (CACE)	A mixture model is used to identify those in the CON group that are likely to have been fully adherent had they been assigned to EXP. Outcomes in this subgroup are compared with outcomes of those who were fully adherent in the EXP group.	1 (4)	None stated	None stated
Test statistic based on the OR in CRTs (estimand unclear)	A test statistic for assessing non-inferiority based on the OR in CRTs under the Dirichlet multinomial model	1 (4)	None stated	None stated
CACE analysis (CACE)	Exact method not stated	1 (4)	None stated	None stated

† Intercurrent event component of the estimand; * As stated by the authors; † both applications of the tipping point approach were reported in the same methodology paper; IV = instrumental variable; CACE = complier average causal effect; ATE = average treatment effect in the population; EXP = experimental intervention; CON = control; OR = odds ratio; CRT = cluster randomised trial.

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Table 4. Details of the statistical methods reported on more than one occasion

Method	Description	Key assumptions	Advantages	Disadvantages
IV approaches[23, 31]	With two randomised groups (*Z*; *Z*=0 for those in the CON group, *Z*=1 for those in the EXP group), a binary measure of the intervention received (*X*; *X*=1 when EXP received, *X*=0 when CON received), and a continuous outcome (*Y*), the conventional IV estimator of CACE is: $IV = \frac{E[Y \mid Z = 1] - E[Y \mid Z = 0]}{\Pr(X = 1 \mid Z = 1) - \Pr(X = 1 \mid Z = 0)}$ Alternatively, 2SLS regression can be used in the presence of confounding by baseline covariates (*C*). In the first stage, *X* is regressed on *Z* and *C.* In the second stage, *Y* is regressed on both the predicted value of *X* obtained from the first stage and *C*. The coefficient for the effect of *X* on *Y* in the second stage provides an estimate of CACE.	The instrument (randomisation): 1) Affects the outcome only through the intervention received (the *exclusion restriction*) 2) Does not share common causes with the outcome (the *exchangeability* assumption) 3) Causes some participants to receive their assigned intervention (the *relevance* assumption) Additional assumption required to estimate CACE: 4) There are no participants who would always receive the opposite of their random allocation (the *monotonicity* assumption)	1) Preserves randomised comparison[30] 2) Assumptions well suited to double-blinded trials (typically assumptions 1 and 4 are satisfied by effective double-blinding and use of objective outcomes, and assumption 2 valid due to randomisation)[32] 3) Under assumptions 1-4, able to estimate CACE even when unmeasured confounding is present[32] 4) Inclusion of confounders in 2SLS regression can improve precision[33] 5) Can be extended to allow for partial adherence, binary or time-to-event outcomes, and clustering, though additional assumptions may be required[36, 46-49]	1) Requires untestable assumptions (1 and 4) that may be violated. When adherence is binary, the exclusion restriction implies no effect of EXP in those who are non-adherent. 2) Only appropriate when crossovers occur and cannot be used when non-trial interventions are received[5] 3) The sample size required to maintain statistical power to detect non-inferiority increases as the quantity of non-adherence increases[5] 4) Simple approaches described involve a single measure of the intervention received (e.g. ≥80% of sessions attended versus <80%, or the proportion of sessions attended) and are susceptible to time-varying confounding (where predictors of both adherence and outcome vary over time)
Adjustment for observed adherence	Observed adherence at a fixed timepoint (e.g. whether surgery was performed) or multiple timepoints (e.g. whether medication was taken adequately between follow-up visits) included as a covariate within a regression model	1) Individuals in the EXP and CON groups with the same level of observed adherence are comparable 2) Within each trial arm, the effect of adherence on the outcome is the same	1) Relatively straightforward to implement	1) Susceptible to selection bias and should be avoided. Different factors may lead to non-adherence in the two groups, meaning those in the EXP group with an observed level of adherence may differ from those in the CON group with the same level of

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Method	Description	Key assumptions	Advantages	Disadvantages
		3) The functional form of adherence is correctly specified 4) Other model assumptions are not violated		adherence. Also, some of those in the CON group that were non-adherent may have been fully adherent if assigned to the EXP group (and vice versa).[33] 2) Does not account for time-varying confounding
Adherence modelled as a time-varying covariate in a time-to-event analysis[50]	An extension of the Cox PH model that allows the intervention received to vary over time. The model takes the form: $$\lambda_i(t) = \lambda_0(t)\exp(\beta X_i(t))$$ where $\lambda_i(t)$ is the hazard function at time t, $\lambda_0(t)$ is the baseline hazard, and $X_i(t)$ takes the value 0 while a participant receives CON and 1 while they receive EXP. β is the log hazard ratio for the effect of receiving EXP versus receiving CON.	1) Only the current value of the intervention (at time t) affects the hazard 2) The effect of receiving EXP is the same for all participants regardless of when it is received 3) Other assumptions of the Cox PH model (including uninformative censoring) are not violated	1) Allows for time-varying confounding that is not influenced by previous intervention received 2) Can be extended to allow for more flexible measures of the intervention received, such as cumulative exposure to the intervention 3) Non-trial interventions may be incorporated	1) Assumptions 1 and 2 are difficult to verify and may be violated 2) Susceptible to selection bias if switching is related to prognostic factors[50] 3) Does not account for time-varying confounding that is influenced by previous intervention received
Tipping point approach[43]	Outcome data following non-adherence is treated as missing. A range of assumptions about these outcomes are explored to assess how sensitive the trial results are to the missing values.	Assumptions made about the values of missing outcomes following non-adherence e.g. all missing outcomes are, 1) failures, 2) successes in the CON group and failures in the EXP group (worst case scenario), or 3) failures in the CON group and successes in the EXP group (best case scenario)	1) All randomised individuals are included in the analysis[43] 2) A model or mechanism for the missing outcomes does not have to be assumed[43]	1) While a range of assumptions about the missing values can be explored, these assumptions are often difficult or not possible to verify[43]
Rank preserving structural failure time model and G-	Let T_i denote the observed survival time for the ith participant and U_i their survival time that would have been observed if they received no EXP. T_i is assumed to be a function of time on (	1) If no participants received any EXP, the average survival time in the two groups would be equal (due to randomisation)	1) Preserves randomised comparison[45] 2) Takes adherence history into account[44]	1) Assumption 2 and 3 are difficult to verify and may be violated[50] 2) Additional assumptions required when the CON group receive an active intervention[51]

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Method	Description	Key assumptions	Advantages	Disadvantages
estimation[39, 50, 51]	T_{ON_i}) and time off (T_{OFF_i}) EXP and T_i related to U_i by the causal model: $$U_i = T_{OFF_i} + e^{-\psi_0} T_{ON_i}$$ e^{ψ_0} is the amount by which expected survival times are increased by EXP (the *acceleration factor*). Due to randomisation, U_i is assumed to be independent of trial arm. Untreated event times are predicted for all participants and the value of ψ_0 that results in equal untreated survival times in the randomised groups identified (using G-estimation). This value of ψ_0 is used to calculate adjusted survival times that would have been observed had switching not occurred.	2) The effect of receiving EXP is the same for all participants regardless of when it is received 3) If participant *i* experiences the event of interest before participant *j* when both are treated, then participant *i* would also experience the event before participant *j* when both are untreated (*rank preserving*)	3) Allows for time-varying confounding, including when these confounders are affected by previous intervention received 4) Does not require information on potential confounders (only randomised group, observed event times, and intervention history)[52]	3) Only appropriate when crossovers occur and cannot be used when non-trial interventions are received [52] 4) G-estimation may not work well if the number of participants or events is small[51] 5) Computationally intensive[53]
Inverse-probability-of-treatment weighting[24, 53-55]	Confounding is accounted for by re-weighting participants' outcomes. Typically, logistic regression is used to predict probabilities of adherence at each visit given previous values of the intervention received and confounders. The inverse of these probabilities (the weights) are used to create a pseudo-population in which time-varying confounders are not associated with the intervention. The causal effect of EXP is estimated by performing an unadjusted analysis in the pseudo-population, which is equivalent to a weighted analysis in the original cohort.	1) The absence of unmeasured confounding (*exchangeability*) 2) Participants have a non-zero probability of receiving each intervention (*positivity*) 3) The observed outcome equals the counterfactual outcome of the intervention actually received (*consistency*) 4) No misspecification of the model used to estimate the weights	1) Preserves randomised comparison[11] 2) Takes adherence history into account 3) Allows for time-varying confounding, including when these confounders are affected by previous intervention received 4) Can be extended to allow for non-trial interventions, but may result in large weights and estimates very sensitive to model specification[54]	1) Eliminates bias if all confounders can be appropriately adjusted for, but in general this will not be possible[11] 2) Cannot be used if covariates perfectly predict adherence[53] 3) Unstable in the presence of extreme weights[53]

IV = instrumental variable; CON = control; EXP = experimental intervention; CACE = complier average causal effect; 2SLS = two-stage least squares; PH = proportional hazards.

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REFERENCES

1. Piaggio G, Elbourne DR, Altman DG, Pocock SJ, Evans SJ. Reporting of noninferiority and equivalence randomized trials: an extension of the CONSORT statement. Jama. 2006 Mar 8;295(10):1152-60
2. Suda KJ, Hurley AM, McKibbin T, Motl Moroney SE. Publication of noninferiority clinical trials: changes over a 20-year interval. Pharmacotherapy. 2011 Sep;31(9):833-9
3. Bikdeli B, Welsh JW, Akram Y, Punnanithinont N, Lee I, Desai NR, et al. Noninferiority Designed Cardiovascular Trials in Highest-Impact Journals. Circulation. 2019 Jul 30;140(5):379-89
4. Piaggio G, Elbourne DR, Pocock SJ, Evans SJ, Altman DG. Reporting of noninferiority and equivalence randomized trials: extension of the CONSORT 2010 statement. Jama. 2012 Dec 26;308(24):2594-604
5. Mo Y, Lim C, Watson JA, White NJ, Cooper BS. Non-adherence in non-inferiority trials: pitfalls and recommendations. Bmj. 2020 Jul 1;370:m2215
6. Dodd S, White IR, Williamson P. Nonadherence to treatment protocol in published randomised controlled trials: a review. Trials. 2012 Jun 18;13:84
7. DiMatteo MR. Variations in patients' adherence to medical recommendations: a quantitative review of 50 years of research. Med Care. 2004 Mar;42(3):200-9
8. DiMatteo MR, Giordani PJ, Lepper HS, Croghan TW. Patient adherence and medical treatment outcomes: a meta-analysis. Med Care. 2002 Sep;40(9):794-811
9. Simpson SH, Eurich DT, Majumdar SR, Padwal RS, Tsuyuki RT, Varney J, et al. A meta-analysis of the association between adherence to drug therapy and mortality. Bmj. 2006 Jul 1;333(7557):15
10. Ye C, Beyene J, Browne G, Thabane L. Estimating treatment effects in randomised controlled trials with non-compliance: a simulation study. BMJ Open. 2014 Jun 17;4(6):e005362
11. Mo Y, Lim C, Mukaka M, Cooper BS. Statistical considerations in the design and analysis of non-inferiority trials with binary endpoints in the presence of non-adherence: a simulation study. Wellcome Open Res. 2019;4:207
12. Committee for Proprietary Medicinal Products (CHMP). Points to consider on switching between superiority and non-inferiority. Br J Clin Pharmacol. 2001 Sep;52(3):223-8
13. Rehal S, Morris TP, Fielding K, Carpenter JR, Phillips PP. Non-inferiority trials: are they inferior? A systematic review of reporting in major medical journals. BMJ Open. 2016 Oct 7;6(10):e012594
14. Matilde Sanchez M, Chen X. Choosing the analysis population in non-inferiority studies: per protocol or intent-to-treat. Stat Med. 2006 Apr 15;25(7):1169-81
15. Brittain E, Lin D. A comparison of intent-to-treat and per-protocol results in antibiotic non-inferiority trials. Stat Med. 2005 Jan 15;24(1):1-10
16. Clarke PS, Windmeijer F. Identification of causal effects on binary outcomes using structural mean models. Biostatistics. 2010 Oct;11(4):756-70
17. ICH Harmonised Tripartite Guideline. Statistical principles for clinical trials. International Conference on Harmonisation E9 Expert Working Group. Stat Med. 1999 Aug 15;18(15):1905-42
18. Gupta SK. Intention-to-treat concept: A review. Perspect Clin Res. 2011 Jul;2(3):109-12
19. Nagelkerke N, Fidler V, Bernsen R, Borgdorff M. Estimating treatment effects in randomized clinical trials in the presence of non-compliance. Stat Med. 2000 Jul 30;19(14):1849-64
20. Abraha I, Montedori A. Modified intention to treat reporting in randomised controlled trials: systematic review. BMJ. 2010;340:c2697
21. Hernán MA, Robins JM. Per-Protocol Analyses of Pragmatic Trials. N Engl J Med. 2017 Oct 5;377(14):1391-8
22. DeMets DL. Statistical issues in interpreting clinical trials. J Intern Med. 2004 May;255(5):529-37
23. Angrist JD, Imbens GW, Rubin DB. Identification of Causal Effects Using Instrumental Variables. Journal of the American Statistical Association. 1996;91(434):444-55

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24. Robins JM, Hernán MA, Brumback B. Marginal structural models and causal inference in epidemiology. Epidemiology. 2000 Sep;11(5):550-60
25. Yau LHY, Little RJ. Inference for the Complier-Average Causal Effect From Longitudinal Data Subject to Noncompliance and Missing Data, With Application to a Job Training Assessment for the Unemployed. Journal of the American Statistical Association. 2001 2001/12/01;96(456):1232-44
26. Page MJ, Moher D, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. PRISMA 2020 explanation and elaboration: updated guidance and exemplars for reporting systematic reviews. BMJ. 2021;372:n160
27. Popay J, Roberts H, Sowden A, Petticrew M, Arai L, Rodgers M, et al. Guidance on the conduct of narrative synthesis in systematic reviews: a product of the ESRC methods programme (Version I). Lancaster, UK: University of Lancaster, 2006.
28. Mostazir M, Taylor RS, Henley W, Watkins E. An overview of statistical methods for handling nonadherence to intervention protocol in randomized control trials: a methodological review. J Clin Epidemiol. 2019 Apr;108:121-31
29. Alshreef A, Latimer N, Tappenden P, Wong R, Hughes D, Fotheringham J, et al. Statistical Methods for Adjusting Estimates of Treatment Effectiveness for Patient Nonadherence in the Context of Time-to-Event Outcomes and Health Technology Assessment: A Systematic Review of Methodological Papers. Med Decis Making. 2019 Nov;39(8):910-25
30. Kim MY. Using the instrumental variables estimator to analyze noninferiority trials with noncompliance. J Biopharm Stat. 2010 Jul;20(4):745-58
31. Dunn G, Maracy M, Tomenson B. Estimating treatment effects from randomized clinical trials with noncompliance and loss to follow-up: the role of instrumental variable methods. Stat Methods Med Res. 2005 Aug;14(4):369-95
32. Lousdal ML. An introduction to instrumental variable assumptions, validation and estimation. Emerg Themes Epidemiol. 2018;15:1
33. Stuart EA, Perry DF, Le HN, Ialongo NS. Estimating intervention effects of prevention programs: accounting for noncompliance. Prev Sci. 2008 Dec;9(4):288-98
34. Davies NM, Smith GD, Windmeijer F, Martin RM. Issues in the reporting and conduct of instrumental variable studies: a systematic review. Epidemiology. 2013 May;24(3):363-9
35. Gillespie D, Farewell D, Barrett-Lee P, Casbard A, Hawthorne AB, Hurt C, et al. The use of randomisation-based efficacy estimators in non-inferiority trials. Trials. 2017 Mar 9;18(1):117
36. Angrist JD, Imbens GW. Two-Stage Least Squares Estimation of Average Causal Effects in Models with Variable Treatment Intensity. Journal of the American Statistical Association. 1995 1995/06/01;90(430):431-42
37. Baiocchi M, Cheng J, Small DS. Instrumental variable methods for causal inference. Stat Med. 2014 Jun 15;33(13):2297-340
38. Daniel RM, Cousens SN, De Stavola BL, Kenward MG, Sterne JA. Methods for dealing with time-dependent confounding. Stat Med. 2013 Apr 30;32(9):1584-618
39. Hernán MA, Cole SR, Margolick J, Cohen M, Robins JM. Structural accelerated failure time models for survival analysis in studies with time-varying treatments. Pharmacoepidemiol Drug Saf. 2005 Jul;14(7):477-91
40. Lam WY, Fresco P. Medication Adherence Measures: An Overview. BioMed Research International. 2015 2015/10/11;2015:217047
41. Kitchener HC, Dunn G, Lawton V, Reid F, Nelson L, Smith AR. Laparoscopic versus open colposuspension--results of a prospective randomised controlled trial. Bjog. 2006 Sep;113(9):1007-13
42. Brunori G, Viola BF, Parrinello G, De Biase V, Como G, Franco V, et al. Efficacy and safety of a very-low-protein diet when postponing dialysis in the elderly: a prospective randomized multicenter controlled study. Am J Kidney Dis. 2007 May;49(5):569-80
43. Kim M, Wang C, Xue X. Assessing the influence of treatment nonadherence on noninferiority trials using the tipping point approach. Stat Med. 2019 Feb 20;38(4):650-9

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44. Wu Y, Zhao L, Hou Y, Li K, Zhou X. Correcting for non-compliance in randomized non-inferiority trials with active and placebo control using structural models. Stat Med. 2015 Mar 15;34(6):950-65
45. Matsuyama Y. A comparison of the results of intent-to-treat, per-protocol, and g-estimation in the presence of non-random treatment changes in a time-to-event non-inferiority trial. Stat Med. 2010 Sep 10;29(20):2107-16
46. Nagelkerke N, Fidler V, Bernsen R, Borgdorff M. Estimating treatment effects in randomized clinical trials in the presence of non-compliance. Statistics in Medicine. 2000;19(14):1849-64
47. Palmer TM, Thompson JR, Tobin MD, Sheehan NA, Burton PR. Adjusting for bias and unmeasured confounding in Mendelian randomization studies with binary responses. International Journal of Epidemiology. 2008;37(5):1161-8
48. Tchetgen Tchetgen EJ, Walter S, Vansteelandt S, Martinussen T, Glymour M. Instrumental variable estimation in a survival context. Epidemiology. 2015 May;26(3):402-10
49. Agbla SC, De Stavola B, DiazOrdaz K. Estimating cluster-level local average treatment effects in cluster randomised trials with non-adherence. Stat Methods Med Res. 2020 Mar;29(3):911-33
50. Morden JP, Lambert PC, Latimer N, Abrams KR, Wailoo AJ. Assessing methods for dealing with treatment switching in randomised controlled trials: a simulation study. BMC Med Res Methodol. 2011 Jan 11;11:4
51. Latimer NR, Henshall C, Siebert U, Bell H. TREATMENT SWITCHING: STATISTICAL AND DECISION-MAKING CHALLENGES AND APPROACHES. International Journal of Technology Assessment in Health Care. 2016;32(3):160-6
52. Allison A, White IR, Bond S. rpsftm: An R Package for Rank Preserving Structural Failure Time Models. R j. 2017 Dec 4;9(2):342-53
53. Mansournia MA, Etminan M, Danaei G, Kaufman JS, Collins G. Handling time varying confounding in observational research. BMJ. 2017;359:j4587
54. Ellis AR, Brookhart MA. Approaches to inverse-probability-of-treatment–weighted estimation with concurrent treatments. Journal of Clinical Epidemiology. 2013 2013/08/01/;66(8, Supplement):S51-S6
55. Austin PC, Stuart EA. Moving towards best practice when using inverse probability of treatment weighting (IPTW) using the propensity score to estimate causal treatment effects in observational studies. Stat Med. 2015 Dec 10;34(28):3661-79

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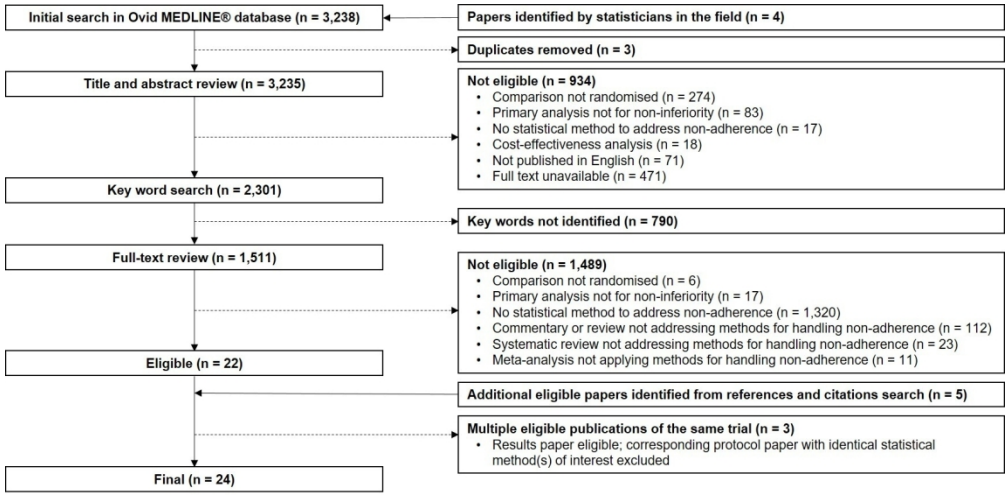


Figure 1. Flow chart showing the eligibility of papers reviewed

146x71mm (300 x 300 DPI)

APPENDIX

The Ovid MEDLINE® database was searched for terms related to treatment adherence, statistical methods for handling non-adherence, and non-inferiority trials in the titles, abstracts and key words of articles published up to 31st December 2020. The full search strategy was as follows:

1. ("intention to treat" or "intention-to-treat" or "ITT" or "intent-to-treat" or "intent to treat").ti,ab,kw.
2. ("as treated" or "as-treated").ti,ab,kw.
3. ("per protocol" or "per-protocol").ti,ab,kw.
4. ("complier average causal effect" or "CACE" or "local average treatment effect").ti,ab,kw.
5. ("non-compliance" or "noncompliance" or "compliance" or "non-adherence" or "nonadherence" or "adherence").ti,ab,kw.
6. ("instrumental variable" or "instrumental variables" or "IV analysis" or "IV analyses").ti,ab,kw.
7. ("non-inferiority" or "noninferiority" or "non-inferior" or "noninferior").ti,ab,kw.
8. 1 or 2 or 3 or 4 or 5 or 6
9. 7 and 8

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Table A1. Characteristics of eligible analyses (n=26)

Characteristics	n (%)
Journal (n=24)	
The New England Journal of Medicine	4 (17)
Statistics in Medicine	3 (13)
BJOG	2 (8)
Journal of Biopharmaceutical Statistics	2 (8)
Trials	2 (8)
Addiction	1 (4)
American Journal of Kidney Disease	1 (4)
BMC Psychiatry	1 (4)
BMJ Open	1 (4)
Clinical Gastroenterology and Hepatology	1 (4)
Clinical Pediatrics	1 (4)
Food and Nutrition Bulletin	1 (4)
The Journal of the American Medical Association (JAMA)	1 (4)
JAMA Network Open	1 (4)
PLOS Medicine	1 (4)
Wellcome Open Research	1 (4)

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Table A2. Explanation of formulae included in Table 4

Method	Formula	Explanation of formula
IV approaches	With two randomised groups (*Z*; *Z*=0 for those in the CON group, *Z*=1 for those in the EXP group), a binary measure of the intervention actually received (*X*; *X*=1 when EXP received, *X*=0 when CON received), and a continuous outcome (*Y*), the conventional IV estimator of CACE is: $$IV = \frac{E[Y\|Z=1] - E[Y\|Z=0]}{\Pr(X=1\|Z=1) - \Pr(X=1\|Z=0)}$$	The conventional IV estimator is equivalent to the ITT effect of *Z* on *Y* (i.e. the ITT effect of randomised group on the outcome) divided by the ITT effect of *Z* on *X* (i.e. the ITT effect of randomised group on the intervention actually received). In other words, the ITT effect of randomised group on the outcome is inflated according to the level of adherence with the randomly assigned interventions in the ITT population. For instance, if the ITT effect of randomised group on the outcome was estimated to be +2% as a risk difference, 95% of those in the CON group received CON, 5% of those in the CON group received EXP, 85% of those in the EXP group received EXP, and 15% of those in the EXP group received CON, the conventional IV estimator of CACE would be: $$IV = \frac{2}{0.85 - 0.05} = 2.5\%$$
Adherence modelled as a time-varying covariate in a time-to-event analysis	An extension of the Cox proportional hazards model that allows the intervention received to vary over time. The model takes the form: $$\lambda_i(t) = \lambda_0(t)\exp(\beta X_i(t))$$ where $\lambda_i(t)$ is the hazard function at time t, $\lambda_0(t)$ is the baseline hazard, and $X_i(t)$ takes the value 0 while a participant receives CON and 1 while they receive EXP. β is the log hazard ratio for the effect of receiving EXP versus receiving CON.	The Cox proportional hazards model is expressed in terms of the hazard function, which is the probability of an outcome occurring at a particular point in time, given that it has not already occurred. The hazard function consists of the baseline hazard (the value of the hazard function when all of the covariates in the model are equal to zero) and the coefficients, which estimate the effects of the covariates. The baseline hazard is able to vary over time and it is assumed that each covariate has a multiplicative effect on the baseline hazard that is constant over time (the proportional hazards assumption). These multiplicative effects are known as hazard ratios. The standard formulation of the Cox model fixes the values of the covariates to be constant over time, but this extension of the method allows the values of the covariates to vary over time. The approach described includes the intervention actually received as the only covariate in the model, the value of which is allowed to vary over time for each participant. If a participant was to

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		switch from receiving CON to EXP, their time at risk and outcome status prior to the switch would be attributed to receiving CON, whereas their time at risk and outcome status following the switch would be attributed to receiving EXP (and vice versa). This allows us to estimate the multiplicative effect of receiving EXP compared with receiving CON on the hazard of the outcome.
Rank preserving structural failure time model and G-estimation	Let T_i denote the observed survival time for the ith participant and U_i their survival time that would have been observed if they received no EXP. T_i is assumed to be a function of time on (T_{ON_i}) and time off (T_{OFF_i}) EXP and T_i related to U_i by the causal model: $$U_i = T_{OFF_i} + e^{-\psi_0} T_{ON_i}$$ e^{ψ_0} is the amount by which expected survival times are increased by EXP (the *acceleration factor*). Due to randomisation, U_i is assumed to be independent of trial arm. Untreated event times are predicted for all participants and the value of ψ_0 that results in equal untreated survival times in the randomised groups identified (using G-estimation). This value of ψ_0 is used to calculate adjusted survival times that would have been observed had switching not occurred.	Rank preserving structural failure time models are used to account for treatment switching in randomised trials with survival outcomes. The method assumes that each participant's counterfactual event time (i.e. the time to the event which would be observed if they received no EXP) is equal to the amount of time that they do not receive EXP, plus the amount of time that they receive EXP multiplied by the effect of receiving EXP. Due to randomisation, we expect the counterfactual event times to be balanced between the randomised groups (similar to the way that we expect the distribution of participant characteristics, such as age and sex, to be balanced between the randomised groups). The effect of receiving EXP is estimated using each participant's observed times on and off EXP and then trying different values for the effect of receiving EXP until balance is achieved on the counterfactual event times between the randomised groups. This iterative process is known as G-estimation. The effect of receiving EXP versus CON is expressed as the relative amount by which average event times are increased by EXP (known as the acceleration factor). Therefore, an acceleration factor >1 represents longer event times (on average) when receiving EXP and an acceleration factor <1 represents shorter event times (on average) when receiving EXP. The acceleration factor is used to calculate adjusted event times that would have been observed had switching not occurred.

IV = instrumental variable; CON = control; EXP = experimental intervention; CACE = complier average causal effect; ITT = intention-to-treat.

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Table A3. Comparison of non-inferiority conclusions from results papers applying statistical methods of interest versus an alternate approach

Reference	Method of interest	Was method of interest applied in primary or sensitivity analysis?	Alternate analyses performed	Comparison of non-inferiority conclusions
Geldsetzer et al. 2018[1]	IV approach	Sensitivity	ITT (primary analysis, all randomised patients)	Different measures of effect
Huang et al. 2018[2]	IV approach	Sensitivity. Applied to PP analysis (patients for whom the trial intervention was achieved at all time points).	ITT (primary analysis, not defined)	Consistent – both concluded non-inferiority
Beaver et al. 2017[3]	IV approach	Primary	ITT (not defined), PP (those who had their first post-randomisation appointment in line with the randomisation), and AT (not defined)	Consistent – all concluded non-inferiority
Kitchener et al. 2006[4]	IV approach	Sensitivity	ITT (primary analysis, not defined)	Consistent – both concluded non-inferiority
Halpern et al. 2020[5]	IV approach	Sensitivity	mITT (primary analysis, those who were randomised, were not subsequently found to be ineligible, and did not withdraw consent)	Different measures of effect
Butler et al. 2019[6]	IV approach	Primary	mITT (those who had undergone randomisation and had available outcome data, regardless of protocol deviations or the intervention they received)	Consistent – both concluded non-inferiority
Bilimoria et al. 2016[7]	IV approach	Sensitivity	ITT (primary analysis, not defined), PP (limited to adherent programs), and AT (which assessed actual exposure to policy change)	Different measures of effect

Reference	Method of interest	Was method of interest applied in primary or sensitivity analysis?	Alternate analyses performed	Comparison of non-inferiority conclusions
Rimoin et al. 2011[8]	Adjustment for observed adherence	Not stated. Applied to ITT (all randomised patients meeting the inclusion and exclusion criteria) and PP analyses (those who were adherent and returned for the follow-up visit).	Unadjusted ITT and unadjusted PP	Consistent – all adjusted analyses resulted in the same conclusions as the corresponding unadjusted analyses
Hahn et al. 2019[9]	Adherence modelled as a time-varying covariate in a time-to-event analysis	Sensitivity	ITT (primary analysis, all randomised patients according to original group allocation) and PP (excluded patients who did not receive the assigned intervention)	Different measures of effect
Dignass et al. 2009[10]	Adherence modelled as a time-varying covariate in a time-to-event analysis	Sensitivity	ITT (all randomised patients who received some intervention and had at least 1 post-baseline efficacy assessment), PP1 (those who dropped out of the study censored at the time of dropout) and PP2 (excluded dropouts)	Results not provided in full
Brunori et al. 2007[11]	Adherence modelled as a time-varying covariate in a time-to-event analysis	Sensitivity	ITT (patients were considered part of the diet group even after a switch to dialysis) and PP (diet group patients were censored when switched to dialysis)	Different measures of effect
Flum et al. 2020[12]	Inverse-probability-of-treatment weighting	Sensitivity. Applied to PP analysis (those adherent to the randomisation assignment).	ITT (primary analysis, not defined)	Consistent – both concluded non-inferiority

CACE = complier average causal effect; IV = instrumental variable; ITT = intention-to-treat; PP = per-protocol; AT = as-treated; mITT = modified intention-to-treat.

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REFERENCES

1. Geldsetzer P, Francis JM, Sando D, Asmus G, Lema IA, Mboggo E, et al. Community delivery of antiretroviral drugs: A non-inferiority cluster-randomized pragmatic trial in Dar es Salaam, Tanzania. PLoS Med. 2018 Sep;15(9):e1002659
2. Huang DT, Yealy DM, Filbin MR, Brown AM, Chang CH, Doi Y, et al. Procalcitonin-Guided Use of Antibiotics for Lower Respiratory Tract Infection. N Engl J Med. 2018 Jul 19;379(3):236-49
3. Beaver K, Williamson S, Sutton C, Hollingworth W, Gardner A, Allton B, et al. Comparing hospital and telephone follow-up for patients treated for stage-I endometrial cancer (ENDCAT trial): a randomised, multicentre, non-inferiority trial. Bjog. 2017 Jan;124(1):150-60
4. Kitchener HC, Dunn G, Lawton V, Reid F, Nelson L, Smith AR. Laparoscopic versus open colposuspension--results of a prospective randomised controlled trial. Bjog. 2006 Sep;113(9):1007-13
5. Halpern SD, Small DS, Troxel AB, Cooney E, Bayes B, Chowdhury M, et al. Effect of Default Options in Advance Directives on Hospital-Free Days and Care Choices Among Seriously Ill Patients: A Randomized Clinical Trial. JAMA Netw Open. 2020 Mar 2;3(3):e201742
6. Butler CC, Gillespie D, White P, Bates J, Lowe R, Thomas-Jones E, et al. C-Reactive Protein Testing to Guide Antibiotic Prescribing for COPD Exacerbations. N Engl J Med. 2019 Jul 11;381(2):111-20
7. Bilimoria KY, Chung JW, Hedges LV, Dahlke AR, Love R, Cohen ME, et al. National Cluster-Randomized Trial of Duty-Hour Flexibility in Surgical Training. N Engl J Med. 2016 Feb 25;374(8):713-27
8. Rimoin AW, Hoff NA, Fischer Walker CL, Hamza HS, Vince A, Abdel Rahman N, et al. Treatment of streptococcal pharyngitis with once-daily amoxicillin versus intramuscular benzathine penicillin G in low-resource settings: a randomized controlled trial. Clin Pediatr (Phila). 2011 Jun;50(6):535-42
9. Hahn JY, Song YB, Oh JH, Chun WJ, Park YH, Jang WJ, et al. Effect of P2Y12 Inhibitor Monotherapy vs Dual Antiplatelet Therapy on Cardiovascular Events in Patients Undergoing Percutaneous Coronary Intervention: The SMART-CHOICE Randomized Clinical Trial. Jama. 2019 Jun 25;321(24):2428-37
10. Dignass AU, Bokemeyer B, Adamek H, Mross M, Vinter-Jensen L, Börner N, et al. Mesalamine once daily is more effective than twice daily in patients with quiescent ulcerative colitis. Clin Gastroenterol Hepatol. 2009 Jul;7(7):762-9
11. Brunori G, Viola BF, Parrinello G, De Biase V, Como G, Franco V, et al. Efficacy and safety of a very-low-protein diet when postponing dialysis in the elderly: a prospective randomized multicenter controlled study. Am J Kidney Dis. 2007 May;49(5):569-80
12. Flum DR, Davidson GH, Monsell SE, Shapiro NI, Odom SR, Sanchez SE, et al. A Randomized Trial Comparing Antibiotics with Appendectomy for Appendicitis. N Engl J Med. 2020 Nov 12;383(20):1907-19

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PRISMA 2020 Main Checklist

Topic	No.	Item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Title (page 1), abstract (page 1)
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist	
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Paragraphs 7 and 8 of introduction section (page 6)
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Paragraph 8 of introduction section (page 6)
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Paragraph 1 of methods section (pages 6 and 7)
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Paragraph 1 of methods section (pages 6 and 7)
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Full search strategy provided in the appendix
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Paragraph 1 of methods section (pages 6 and 7)
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Paragraph 1 of methods section (pages 6 and 7)

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Topic	No.	Item	Location where item is reported
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Paragraph 2 of methods section (pages 7 and 8)
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Paragraph 2 of methods section (pages 7 and 8)
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	N/A
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	N/A
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item 5)).	N/A
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	N/A
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	N/A
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Software package (paragraph 3 of methods section on page 8)
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	N/A
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	N/A
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	N/A

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Topic	No.	Item	Location where item is reported
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	N/A
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Paragraph 1 of results section (page 8) and Figure 1 (page 16)
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Paragraph 1 of results section (page 8) and Figure 1 (page 16)
Study characteristics	17	Cite each included study and present its characteristics.	Table 1 (pages 17 and 18)
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	N/A
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	N/A
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	N/A
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	N/A
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	N/A
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	N/A
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	N/A
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	N/A

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Topic	No.	Item	Location where item is reported
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Paragraph 3 of discussion section (page 11)
	23b	Discuss any limitations of the evidence included in the review.	N/A
	23c	Discuss any limitations of the review processes used.	Strengths and limitations section (page 14)
	23d	Discuss implications of the results for practice, policy, and future research.	Paragraph 7 of discussion section (page 13)
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Abstract (page 2) and paragraph 3 of methods section (page 8)
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Abstract (page 2) and paragraph 3 of methods section (page 8)
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Funding section (page 15)
Competing interests	26	Declare any competing interests of review authors.	Competing interests section (page 15)
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Data sharing statement (page 15)

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PRIMSA Abstract Checklist

Topic	No.	Item	Reported?
TITLE			
Title	1	Identify the report as a systematic review.	Yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Yes
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Yes
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	N/A
Synthesis of results	6	Specify the methods used to present and synthesize results.	N/A
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Yes
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Yes
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	No
Interpretation	10	Provide a general interpretation of the results and important implications.	Yes
OTHER			
Funding	11	Specify the primary source of funding for the review.	No
Registration	12	Provide the register name and registration number.	Yes

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